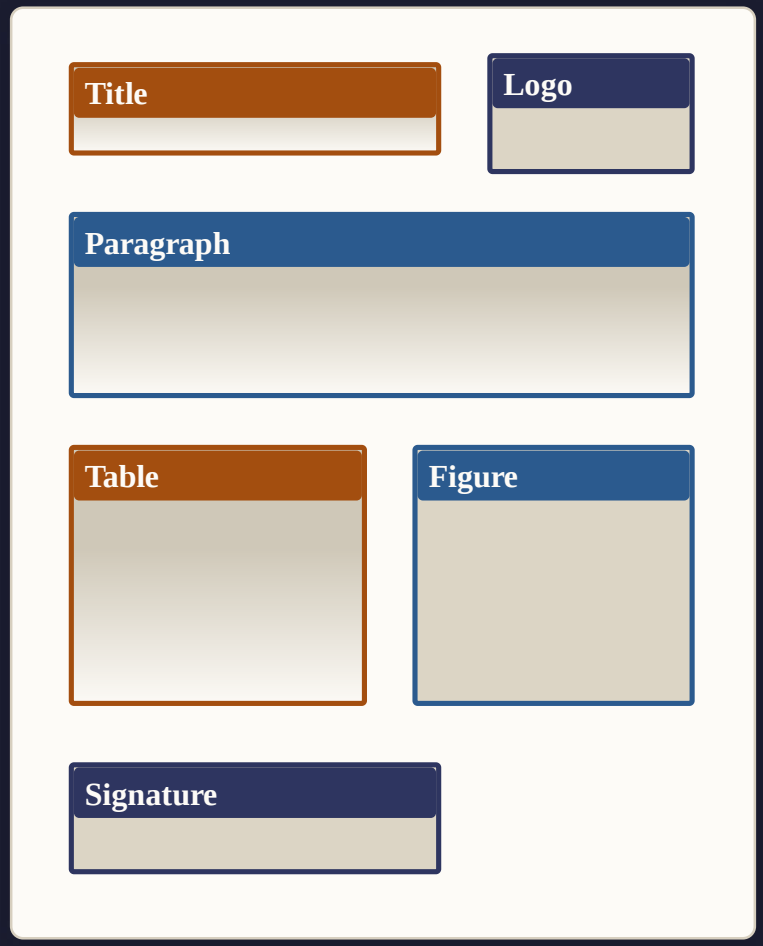
OPTICAL CHARACTER RECOGNITION AND NATURAL LANGUAGE PROCESSING

From pixels on a page to meaning in a sentence

How machines read documents, understand language, and how the two combine into systems that turn paper, scans and photos into structured, searchable, useful knowledge.



BEFORE WE BEGIN

ML and AI engineers

Who build document pipelines, chatbots, search and extraction systems.

Data scientists

Who need to turn scanned and unstructured text into usable data.

Product and operations leads

Who buy, scope or sign off document automation projects.

Students and career changers

Who want a lucid map of OCR and NLP without heavy mathematics.

KEY IDEA

The one idea to hold

OCR turns **images of text into characters**. NLP turns **characters into meaning**. Together they let software read the world's documents.

ANALOGY

OCR is the eyes that recognise letters on a page. NLP is the mind that understands what the letters say. A reader needs both.

BEFORE WE BEGIN

What you will be able to do by the end

- Explain every stage of a modern OCR pipeline, from capture to text
- Choose preprocessing steps that genuinely help recognition
- Describe how text detection and recognition models work
- Extract fields, tables and key values from forms and invoices
- Measure OCR quality with character and word error rates
- Tokenise, normalise and represent text for models
- Explain embeddings, attention and transformers in plain words
- Choose between classical NLP, fine tuned models and LLMs
- Build retrieval augmented systems over scanned documents
- Evaluate, deploy and govern OCR and NLP systems responsibly

HOW TO READ THIS DECK

Every coloured box has a job

Each box type carries a different kind of message. Learn the colour code once and you can skim any slide in seconds.

KEY IDEA

The core concept of the slide. If you read nothing else, read these.

PITFALL

A mistake real teams make, often without noticing until production.

PRO TIP

Practical advice from experience that saves time or rework.

ANALOGY

An everyday comparison that makes an abstract idea concrete.

DEFINITION

A precise meaning for a term, written in plain language.

IN THE WILD

A realistic scenario showing the idea at work in a project.

HOW TO READ THIS DECK

More box types you will meet

MYTH

A popular belief that sounds right but leads you astray.

REALITY

What actually holds true, and why.

RULE OF THUMB

A quick heuristic for when you have no time to think deeply.

CHECK YOURSELF

A short question to test your understanding.

INTERVIEW ANGLE

How the topic tends to come up in technical interviews.

LESSON

The transferable takeaway from a case study.

WARNING SIGNS

Symptoms that tell you something is going wrong.

THE FIX

The remedy for the problem just described.

USE IT WHEN

The situations where a technique is the right choice.

ROADMAP

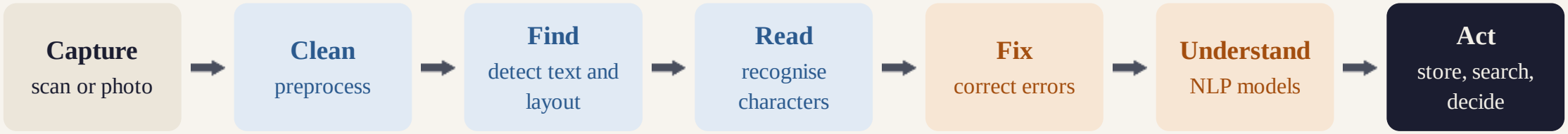
Three parts, twenty one sections

Part A · OCR, sections 1 to 9 | **Part B · NLP, sections 10 to 18** | **Part C · Together, sections 19 to 21**

01 What OCR is	02 Images and capture	03 Preprocessing
04 Layout and text detection	05 Text recognition	06 Postprocessing and correction
07 Document AI	08 Evaluating OCR	09 OCR tools and pipelines
10 NLP foundations	11 Preprocessing and tokenisation	12 Representing text
13 Classical NLP tasks	14 Sequence models	15 Transformers
16 Large language models	17 Applied NLP	18 Evaluating NLP
19 OCR meets NLP	20 Case studies	21 Deployment, ethics and toolkit

THE BIG PICTURE

From a photo of paper to a useful answer



KEY IDEA

The first five boxes are **OCR**: they turn pixels into reliable text. The last two are **NLP**: they turn text into structure and meaning that software can act on.

PITFALL

Errors flow downstream. A misread digit in an invoice total survives every later step unless something checks it. Each stage must hand the next one clean, confident output.

ANALOGY

Think of a postal sorting office. Someone opens the envelope (capture), flattens the letter (clean), finds the address block (find), reads it (read), checks it against a list of real streets (fix), works out which department it is for (understand), and delivers it (act).

THE PREMISE

Reading is two problems: seeing the letters, and understanding the words.

OCR solves the first. NLP solves the second. The most valuable systems in industry today do both, and the seams between them are where most projects succeed or fail.

PART A

Part A · Optical Character Recognition

Nine sections on how machines turn images of text into characters: capture, cleaning, layout, recognition, correction, document understanding, measurement and tools.

01

SECTION 01

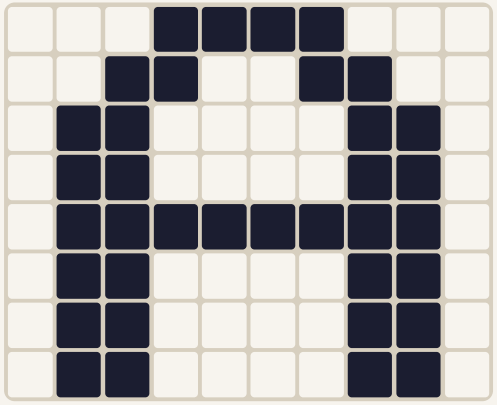
What OCR is

The problem OCR solves, where it came from, the kinds of text it faces, and why it remains hard.

Definition | History | Printed vs handwritten | Scene text | Why it is hard | Use cases

DEFINITION

Optical character recognition in one picture



What the computer receives: a grid of pixel values





What we want: the character A, as text

DEFINITION

OCR is the conversion of images that contain text (scans, photos, screenshots, video frames) into machine readable characters that software can search, edit and analyse.

IN PLAIN WORDS

To a computer an image is only numbers describing brightness. OCR is the bridge from those numbers to letters.

A SHORT HISTORY

Eighty years of teaching machines to read

1910s to 1950s

Reading machines

Early devices read printed characters for telegraphy and to help blind readers.

1970s

Omni font OCR

Systems that read many typefaces, with reading machines for the blind.

1990s

Desktop OCR

Scanners and software bring OCR to offices; engines like Tesseract begin.

2010s

Deep learning

Neural networks read scene text and handwriting far better than rules.

2020s

Document AI

Transformers read layout and text together; vision language models read pages directly.

KEY IDEA

The shift that matters: from **hand written rules** about letter shapes, to models that **learn** what letters look like from millions of examples.

REMEMBER

Old ideas still appear inside modern systems: binarisation, connected components and dictionaries remain useful helpers around neural models.

KINDS OF TEXT

Not all text is equally easy to read

Easy

Clean printed documents

Typed letters, reports, books. Clear fonts, good contrast. The easiest case.

Moderate

Forms and invoices

Printed text in boxes and tables, often with stamps, lines and handwriting mixed in.

Hard

Handwriting

Every writer is different; letters join and vary. Called handwritten text recognition.

Hard

Scene text

Signs, labels, number plates, product packaging photographed in the wild.

Very hard

Historical documents

Faded ink, old fonts, damaged paper, spelling that differs from today.

Easy to moderate

Screenshots and digital images

Rendered text in apps and slides; usually clean but often small.

WHY OCR IS HARD

Ten things that confuse a reading machine

WARNING SIGNS

Poor lighting and shadows across the page

WARNING SIGNS

Blur from camera shake or focus

WARNING SIGNS

Skew and perspective from angled photos

WARNING SIGNS

Low resolution, so letters are a few pixels high

WARNING SIGNS

Noise: speckles, stains, bleed through from the back

WARNING SIGNS

Unusual fonts and decorative lettering

WARNING SIGNS

Touching or broken characters

WARNING SIGNS

Complex layouts: columns, tables, sidebars

WARNING SIGNS

Mixed languages and scripts on one page

WARNING SIGNS

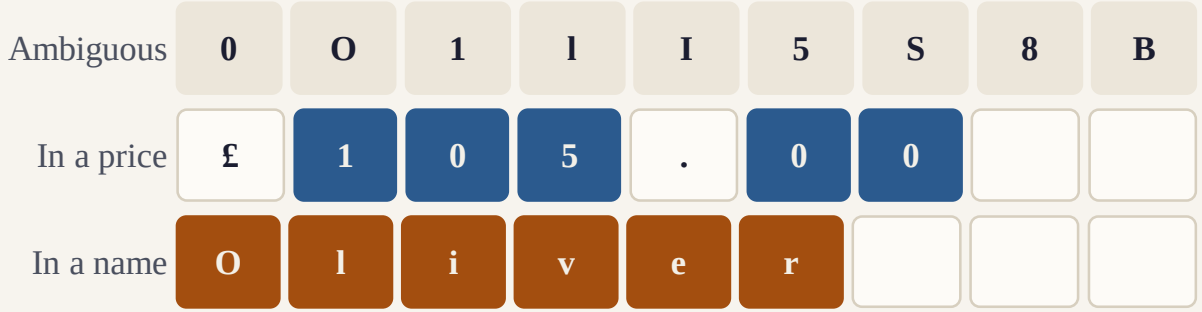
Look alike characters: 0 and O, 1 and l, 5 and S

PRO TIP

When an OCR project struggles, check these ten first. Most failures trace back to capture and layout, not to the recognition model itself.

LOOK ALIKE CHARACTERS

Why context matters even for single letters



KEY IDEA

The same shape can be a digit or a letter. Humans resolve it from **context**: a price contains digits, a name contains letters. Good OCR systems do the same with language models and field rules.

WORKED EXAMPLE

“Invoice tota1: £1O5.0O” is what a shape only reader might produce. A context aware reader returns “Invoice total: £105.00”.

WHERE OCR IS USED

Applications across industries

Sector	Typical use	What success looks like
Finance	Invoices, receipts, bank statements, cheques	Fields extracted accurately with little manual keying
Healthcare	Referral letters, lab reports, prescriptions	Records searchable and coded, with privacy protected
Government	Forms, passports, archives	Faster processing, digitised public records
Logistics	Shipping labels, delivery notes, container codes	Parcels tracked without manual reading
Legal	Contracts, court bundles, discovery	Clauses found and compared in seconds
Retail	Shelf labels, receipts, product packaging	Prices and products matched automatically
Heritage	Newspapers, manuscripts, letters	Centuries of text made searchable

MYTHS AND REALITIES

What people get wrong about OCR

MYTH

OCR is a solved problem.

REALITY

It is solved for clean print. Handwriting, photos and complex layouts remain hard.

MYTH

A better model fixes everything.

REALITY

Capture quality and layout often matter more than the recogniser.

MYTH

Ninety nine per cent character accuracy means the job is done.

REALITY

One wrong character per hundred still means a wrong digit on many invoices. Field accuracy is what counts.

SECTION 1 RECAP

What OCR is

REMEMBER

Takeaways

- OCR turns images of text into characters
- Difficulty depends on text type: print, forms, handwriting, scene text
- Capture and layout cause most failures
- Context resolves look alike characters
- Measure success by the task, not just characters

CHECK YOURSELF

A model reads 99% of characters correctly on invoices with 50 characters in key fields. Is that good enough?

ANSWER

Often not. One error per hundred characters means many invoices will have at least one wrong key field. Measure field level accuracy instead.

SECTION 02

02

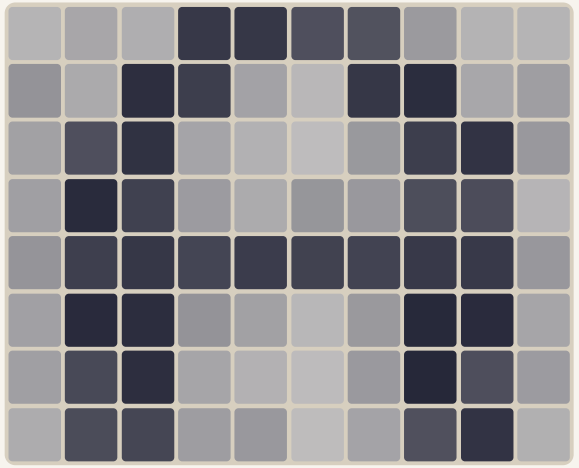
Images and capture

How digital images represent text, why resolution and lighting decide quality, and how to capture documents well.

Pixels **Resolution** **Colour** **Scanners vs cameras** **File formats** **Capture checklist**

PIXELS

An image is a grid of numbers



A greyscale crop of the letter A: each square is one pixel

DEFINITION

A **pixel** is the smallest dot of an image. In greyscale each pixel holds one brightness value, from dark to light. In colour it holds three: red, green and blue.

KEY IDEA

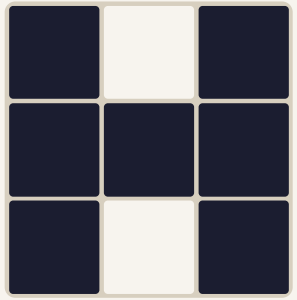
OCR models never see shapes directly. They learn which patterns of brightness values tend to form which characters.

ANALOGY

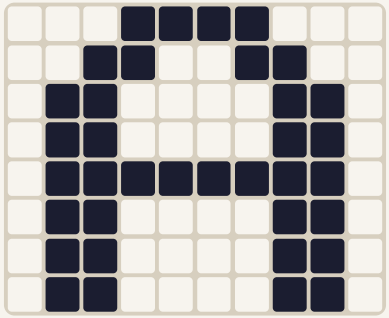
Like a mosaic made of tiny tiles. Stand close and you see only tiles; step back and a letter appears. OCR has to step back.

RESOLUTION

How many pixels does a letter need?



Too few pixels: shape is lost



Enough pixels: shape is clear

DEFINITION

Resolution for scans is measured in dots per inch (DPI): how many pixels cover one inch of paper.

Scan setting	Typical result
150 DPI	Readable for large print only
300 DPI	The usual standard for OCR
400 to 600 DPI	Small print, fine detail, archives

RULE OF THUMB

Aim for body text letters at least about 20 to 30 pixels tall. Below roughly 10 pixels, accuracy drops sharply.

COLOUR AND CONTRAST

Colour, greyscale and black and white

Colour

Three values per pixel. Keeps stamps, highlights and coloured ink distinct. Larger files.

Greyscale

One value per pixel. The default for most OCR. Keeps soft edges that help recognition.

Black and white

Each pixel is ink or paper. Small and fast, but thresholding mistakes are permanent.

PRO TIP

Capture in colour or greyscale and convert later. You can always throw information away, but you can never get it back.

IN THE WILD

A bank captured cheques in pure black and white to save storage. Light blue handwriting vanished entirely on some cheques. Rescanning was impossible because originals had been returned.

CAPTURE DEVICES

Flatbed scanners versus phone cameras

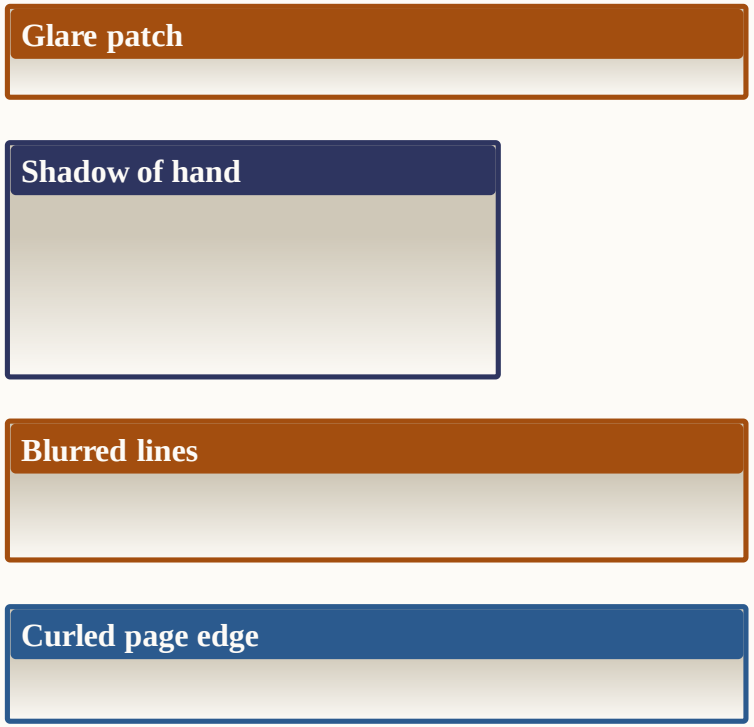
Factor	Flatbed or document scanner	Phone camera
Lighting	Even and controlled	Variable: shadows, glare, colour casts
Geometry	Flat, straight, fixed distance	Tilted, curved pages, perspective distortion
Resolution	Known and consistent DPI	Varies with distance and zoom
Blur	Rare	Common from motion and focus
Convenience	Needs hardware and a desk	Anywhere, by anyone
OCR impact	Highest, most predictable accuracy	Needs strong preprocessing and robust models

KEY IDEA

Most modern OCR products face camera images from customers. Design the pipeline for the **worst** realistic capture, not the best.

CAMERA CAPTURE

What goes wrong with phone photos



WARNING SIGNS

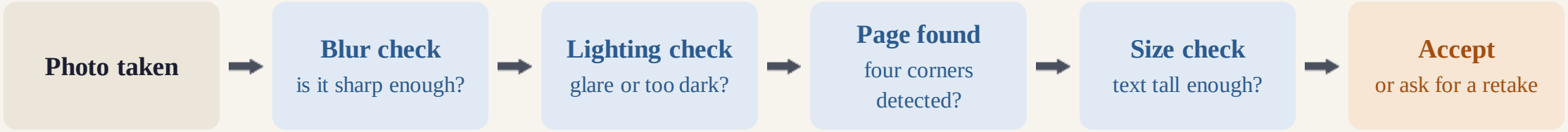
- Glare from glossy paper or plastic sleeves
- Shadows from hands and phones
- Motion blur from shaky hands
- Curved pages near a book spine
- Background clutter around the page

THE FIX

- Guide the user with on screen frames
- Check blur and glare before accepting a photo
- Detect page edges and flatten the page
- Ask for a retake rather than guessing

CAPTURE QUALITY GATES

Reject bad images before OCR, not after



KEY IDEA

A few cheap checks at capture time prevent a large share of OCR errors. Asking for a retake costs the user seconds; fixing a wrong extraction costs far more later.

PRO TIP

A common blur check measures how sharp the edges in the image are. Very soft edges overall mean the photo is blurred. Tune the threshold on your own documents.

FILE FORMATS

Images, PDFs and what is inside them

Format	What it holds	OCR note
PNG	Lossless image	Safe choice for processing; no compression damage
JPEG	Lossy compressed image	Heavy compression creates blocky artefacts around letters
TIFF	Image, often multi page	Common in archives and scanners; supports lossless
PDF with text layer	Real characters plus layout	No OCR needed: extract the text directly
PDF of scans	Images inside a PDF wrapper	Needs OCR on each page image
HEIC and WEBP	Modern phone and web images	Convert carefully before processing

⚠ PITFALL

Many teams run OCR on PDFs that already contain perfect digital text, introducing errors that did not need to exist. Always check for a text layer first.

BORN DIGITAL OR SCANNED?

The first decision in any document pipeline

Incoming PDF

↓

Does it have a usable text layer?

↓

Yes: extract text directly
fast, exact, keeps layout

↓

No or garbled: render pages and run OCR

DEFINITION

Born digital documents were created by software and contain real characters. **Scanned** documents are pictures of paper.

PITFALL

Some PDFs have a hidden text layer made by poor OCR years ago. It looks like text but is full of errors. Sample and check its quality before trusting it.

PRO TIP

Mixed PDFs are common: a typed cover letter followed by scanned attachments. Decide page by page, not file by file.

SECTION 2 RECAP

Images and capture

REMEMBER

Takeaways

- Images are grids of brightness numbers
- About 300 DPI is the usual scanning standard
- Keep greyscale or colour; threshold later
- Phone photos need quality gates and retakes
- Check PDFs for a real text layer before OCR

CHECK YOURSELF

Users upload phone photos of receipts and accuracy is poor. What is the first thing to check?

ANSWER

Capture quality: blur, glare, shadows and text size. Add capture gates and guidance before changing the model.

SECTION 03

03 Preprocessing

Cleaning and normalising images so that recognition sees clear, upright, high contrast text.

Greyscale · Binarisation · Noise removal · Deskew · Perspective · Morphology · Contrast

WHY PREPROCESS

Give the recogniser the easiest possible job

Raw image photo or scan → **Greyscale** one channel → **Denoise** remove speckles → **Deskew** straighten → **Contrast** even out lighting → **Binarise** ink vs paper → **Clean image** ready for OCR

KEY IDEA

Preprocessing removes variation that has nothing to do with the text itself: tilt, shadows, noise, colour casts. The recogniser can then spend its capacity on letter shapes.

PITFALL

Modern neural recognisers are trained on messy images and sometimes do **worse** after heavy preprocessing. Always measure; never assume a step helps.

RULE OF THUMB

Classic engines like Tesseract benefit a lot from clean, binarised input. Deep learning engines usually want light preprocessing: deskew, crop and contrast, then let the model handle the rest.

STEP 1

Converting to greyscale

Colour pixel	Red 210	Green 190	Blue 160
Greyscale pixel	One brightness value		

DEFINITION

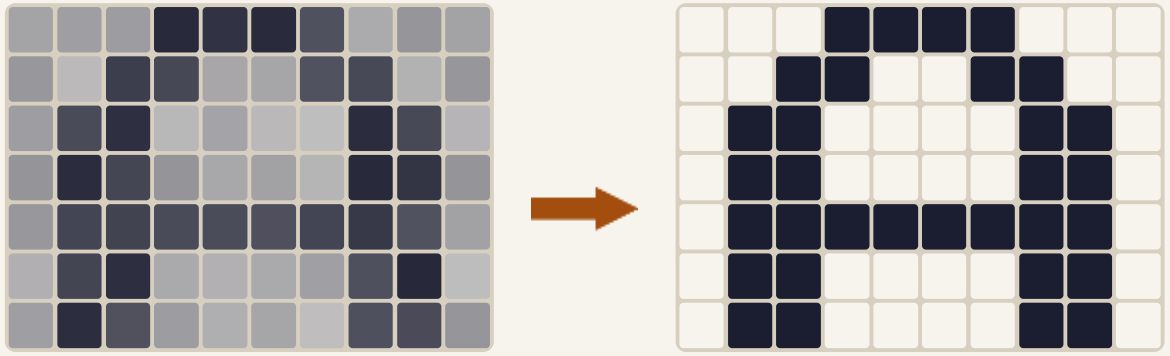
Greyscale conversion combines the three colour channels into a single brightness value per pixel, weighting green most because human eyes are most sensitive to it.

PRO TIP

If text is printed in a coloured ink (red stamps, blue handwriting), try using a **single colour channel** instead. One channel may separate ink from background far better than an average.

STEP 2

Binarisation: deciding ink or paper



Before: shades of grey

After: every pixel is ink or paper

DEFINITION

Binarisation (thresholding) turns each pixel black or white by comparing its brightness with a threshold value.

KEY IDEA

The whole difficulty is choosing the threshold. Too low and faint strokes vanish; too high and the background becomes noise.

STEP 2

Global versus adaptive thresholds

Global threshold

One threshold for the whole image. Otsu’s method chooses it automatically by finding the value that best separates two groups of brightness. Fast and simple.

Adaptive threshold

A different threshold for each small neighbourhood, based on local brightness. Handles shadows and uneven lighting.

Document specific methods

Methods such as Sauvola and Niblack were designed for documents with stains, faded ink and gradients.

WORKED EXAMPLE

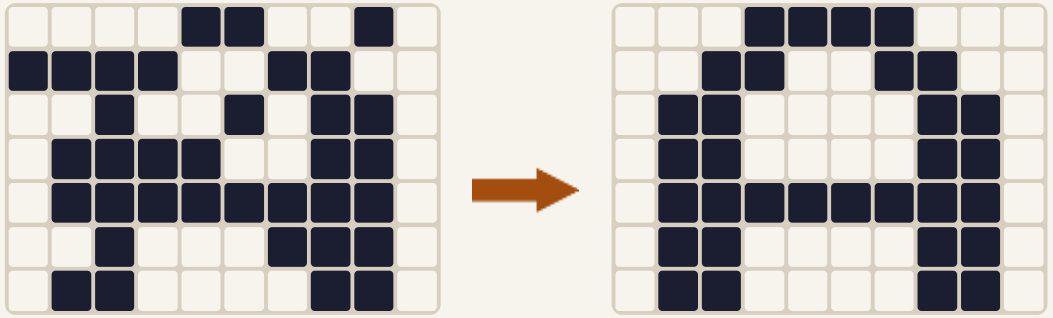
A photo with a shadow across the bottom half: a global threshold turns the shadowed half solid black. Adaptive thresholding keeps the text readable in both halves.

PYTHON · OPENCV

```
import cv2
grey = cv2.imread("page.png", cv2.IMREAD_GRAYSCALE)
flags = cv2.THRESH_BINARY + cv2.THRESH_OTSU
_, otsu = cv2.threshold(grey, 0, 255, flags)
adaptive = cv2.adaptiveThreshold(grey, 255,
    cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
    cv2.THRESH_BINARY, 31, 15)
```

STEP 3

Removing noise



Salt and pepper noise

After a median filter

DEFINITION

Median filter: replaces each pixel with the middle value of its neighbours. Removes speckles, keeps edges.

DEFINITION

Gaussian blur: softens fine noise but also softens letters.

DEFINITION

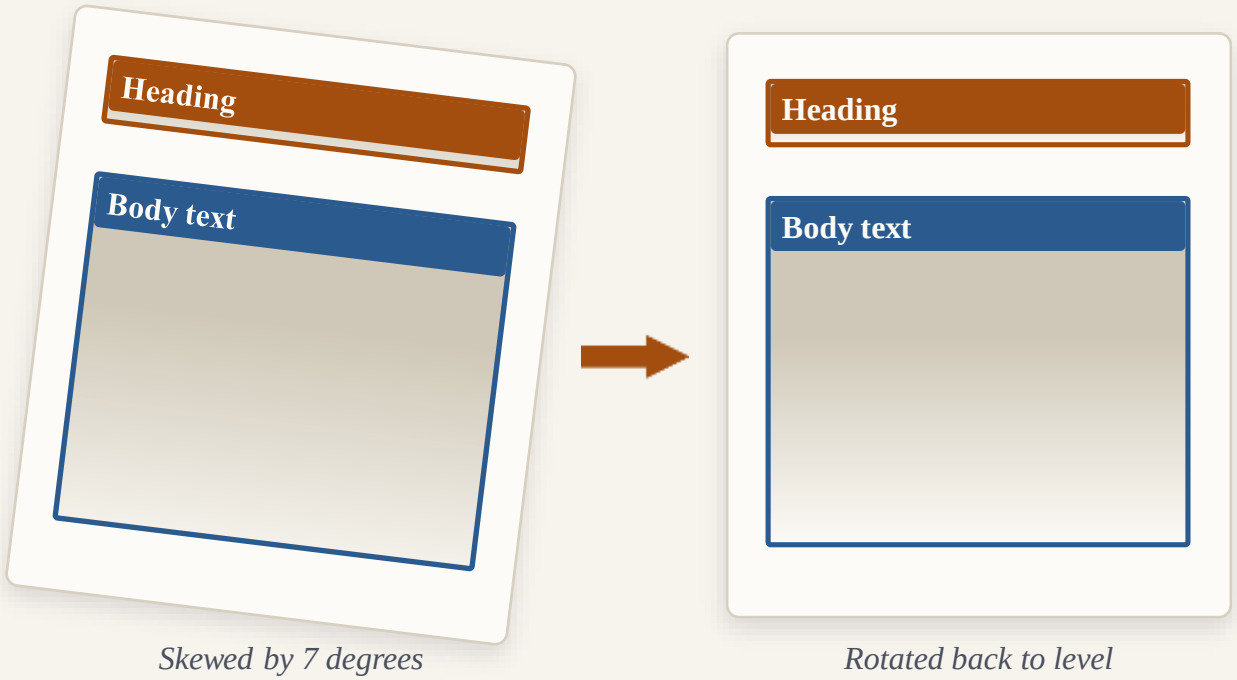
Bilateral filter: smooths flat areas while keeping strong edges.

DEFINITION

Non local means: compares patches across the image for stronger denoising.

STEP 4

Deskewing: straightening the page



Skewed by 7 degrees *Rotated back to level*

DEFINITION

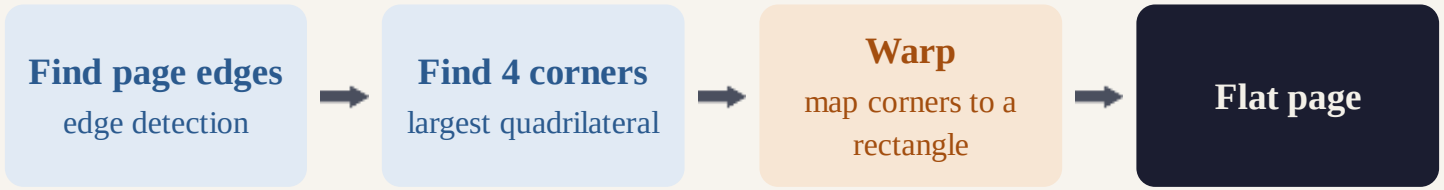
Skew is a small rotation of the text lines. Even a few degrees can confuse line detection and recognition.

KEY IDEA

Common approach: try many small angles and choose the one where the text lines become most clearly separated into horizontal bands.

STEP 5

Perspective correction for photos



DEFINITION

A photo taken at an angle makes the page look like a trapezium. A **perspective transform** stretches the image so the four page corners become a proper rectangle, as if scanned flat.

PITFALL

Curved book pages need more than a four corner warp. Page dewarping models estimate the curve of each line and flatten it.

PRO TIP

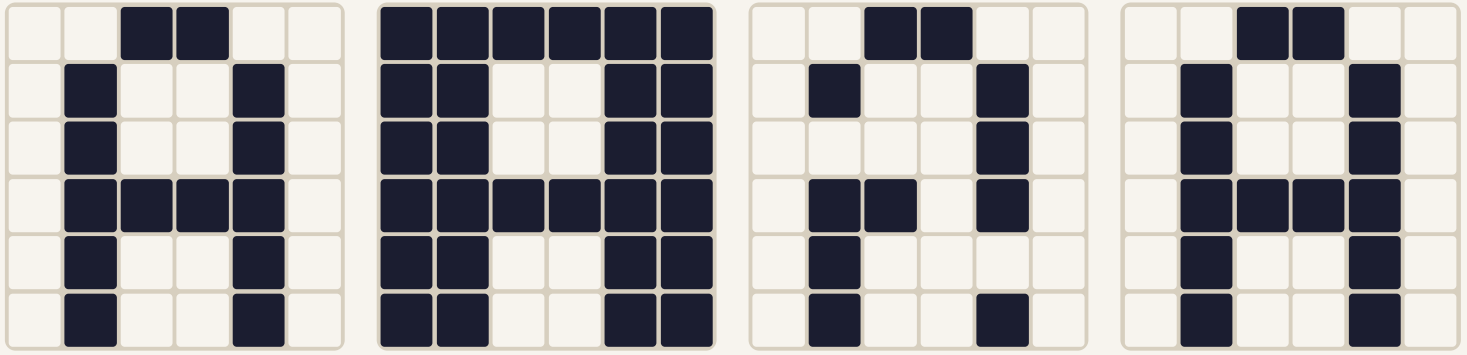
Show the detected corners to the user before capture completes. People fix bad framing instantly when they can see it.

IN THE WILD

A delivery app let drivers photograph signed notes. Adding automatic corner detection and a live outline reduced unreadable uploads dramatically without any model change.

STEP 6

Morphology: thickening, thinning and joining



Thin strokes *After dilation* *Broken strokes* *After closing*

DEFINITION

Dilation grows ink regions. Thickens faint strokes and joins gaps.

DEFINITION

Erosion shrinks ink regions. Separates touching letters and removes specks.

DEFINITION

Closing (dilate then erode) fills small gaps inside strokes.

DEFINITION

Opening (erode then dilate) removes small noise blobs.

STEP 7

Contrast and lighting correction

DEFINITION

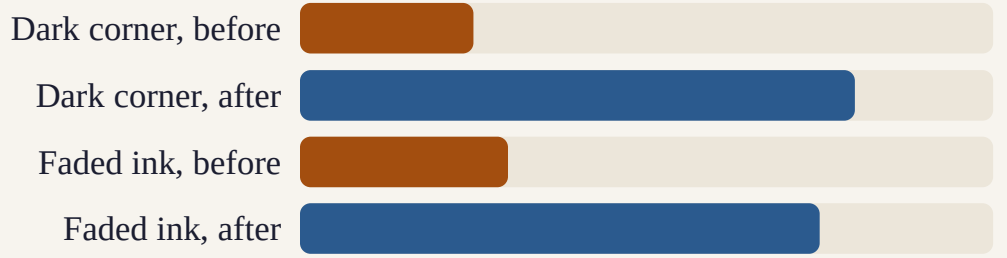
Contrast stretching spreads brightness values across the full range. **Histogram equalisation** redistributes them so details in dark and light areas both show.

DEFINITION

CLAHE equalises contrast in small tiles with a limit on amplification, which avoids blowing up noise.

DEFINITION

Background subtraction estimates the paper colour across the page (for example by heavy blurring) and removes it, leaving only ink.



PITFALL

Strong equalisation also amplifies stains and paper texture. Tune it on a sample of your worst documents.

OTHER STEPS

More preprocessing tools worth knowing

Border removal

Cut black scanner borders and punch holes that look like letters.

Line removal

Remove form lines and table rulings that touch characters.

Rescaling

Resize so text height matches what the model was trained on.

Super resolution

Neural upscaling of tiny text; helps sometimes, invents detail other times.

Watermark removal

Separate light watermarks from foreground ink by colour or intensity.

Orientation detection

Detect pages upside down or sideways before anything else.

CHOOSING STEPS

A preprocessing decision table

Symptom in your images	Try first	Then
Text tilted a few degrees	Deskew	Line based layout analysis
Photographed at an angle	Corner detection and perspective warp	Dewarping for curved pages
Shadows or uneven light	Background subtraction or CLAHE	Adaptive thresholding
Speckles and dust	Median filter	Opening, small blob removal
Faint or broken strokes	Contrast stretch	Closing or dilation
Tiny text	Rescan higher, or upscale	Model trained on small text
Pages rotated 90 or 180 degrees	Orientation classifier	Script and language detection

MEASURE IT

Prove each step earns its place

KEY IDEA

Treat every preprocessing step as a hypothesis. Run OCR with and without it on a representative labelled sample, and keep it only if error rates fall.



WORKED EXAMPLE

Illustrative ablation on a neural recogniser: deskew and contrast helped; aggressive binarisation hurt because it erased faint strokes the model could otherwise read. The bars show relative accuracy, not real benchmark figures.

PRO TIP

Different document types may need different recipes. Route receipts, forms and letters to their own preprocessing chains.

SECTION 3 RECAP

Preprocessing

REMEMBER

Takeaways

- Remove variation unrelated to text
- Adaptive thresholds beat global ones under uneven light
- Deskew and perspective fixes help nearly every engine
- Morphology repairs thin or broken strokes
- Measure every step; neural models may prefer light touch

CHECK YOURSELF

A shadow covers half of every photographed page. Which thresholding approach should you use?

ANSWER

Adaptive thresholding, or background subtraction first. A single global threshold will turn the shadowed half black.

SECTION 04

04 Layout and text detection

Finding where the text is: page regions, reading order, lines, words and tables, with classical and neural detectors.

Page segmentation **Reading order** **Connected components** **Text detectors** **Lines and words** **Tables**

THE QUESTION

Before reading, find what to read



DEFINITION

Layout analysis splits a page into regions (titles, paragraphs, tables, figures, headers, footers) and decides the order in which they should be read.

DEFINITION

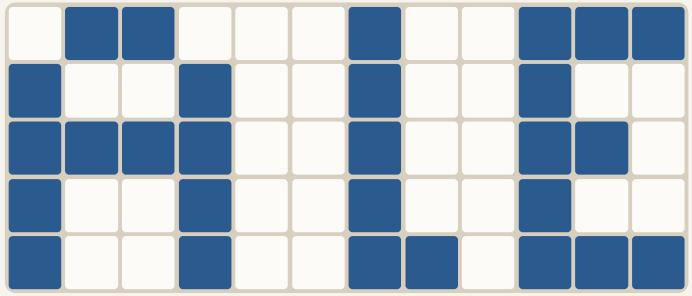
Text detection finds the exact boxes or outlines of text lines or words, so each can be passed to the recogniser.

PITFALL

Reading a two column page straight across mixes sentences from both columns into nonsense. Layout errors look like recognition errors but are not.

CLASSICAL APPROACH

Connected components



Three separate ink blobs

DEFINITION

A **connected component** is a group of ink pixels that all touch each other. On clean print, each letter is usually one component.

KEY IDEA

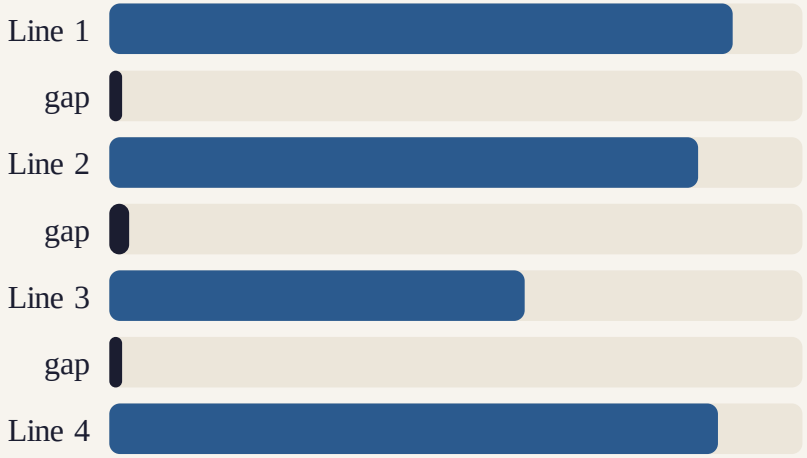
Group components that sit side by side into words, words into lines, lines into blocks. That is the classical bottom up layout method.

PITFALL

Touching letters merge into one component; broken letters split into several; letters like i and j have two parts.

CLASSICAL APPROACH

Projection profiles and whitespace



DEFINITION

A **horizontal projection profile** counts ink pixels in each row. Peaks are text lines; valleys are the gaps between lines.

DEFINITION

Whitespace analysis finds large empty rectangles, which separate columns and blocks.

USE IT WHEN

Clean, upright, printed pages with regular layouts such as books and reports.

CLASSICAL APPROACH

Top down versus bottom up

Top down

Start with the whole page; split it repeatedly at large white gaps (columns, then blocks, then lines). Recursive XY cut is the classic example. Fast; struggles with irregular layouts.

Bottom up

Start with pixels or components; merge them into words, lines and blocks. Handles irregular layouts better; sensitive to noise.

Hybrid and learned

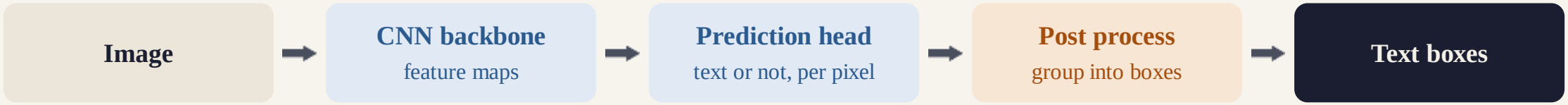
Modern systems learn region detection directly from labelled pages, then apply rules for reading order.

ANALOGY

Top down is cutting a cake into slices along obvious lines. Bottom up is sticking crumbs together until you have slices.

NEURAL TEXT DETECTION

How deep learning finds text



DEFINITION

Regression based: predict box coordinates directly (EAST style).

DEFINITION

Segmentation based: label every pixel as text or background, then group (DB, PSENet style).

DEFINITION

Character region based: find characters and the links between them (CRAFT style).

KEY IDEA

Segmentation based detectors handle curved and oddly shaped text well, which matters for scene text such as logos and signs.

NEURAL TEXT DETECTION

Boxes, rotated boxes and polygons

Simple

Axis aligned box

Four numbers: left, top, width, height. Fine for scanned documents.

Medium

Rotated box

Adds an angle. Handles tilted signs and labels.

Flexible

Polygon

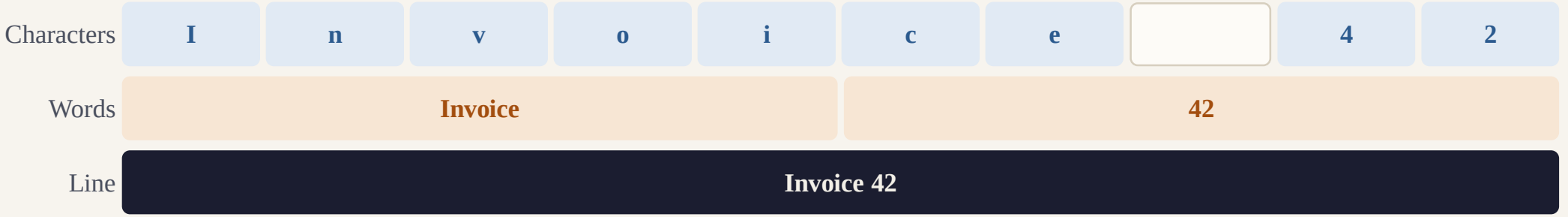
Many points tracing the outline. Handles curved text around logos and bottles.

WORKED EXAMPLE

A coffee cup with a brand name curved around it: an axis aligned box includes lots of background and cuts letters; a polygon follows the curve so the recogniser gets clean, straightened text.

LINES AND WORDS

Choosing the unit you recognise



KEY IDEA

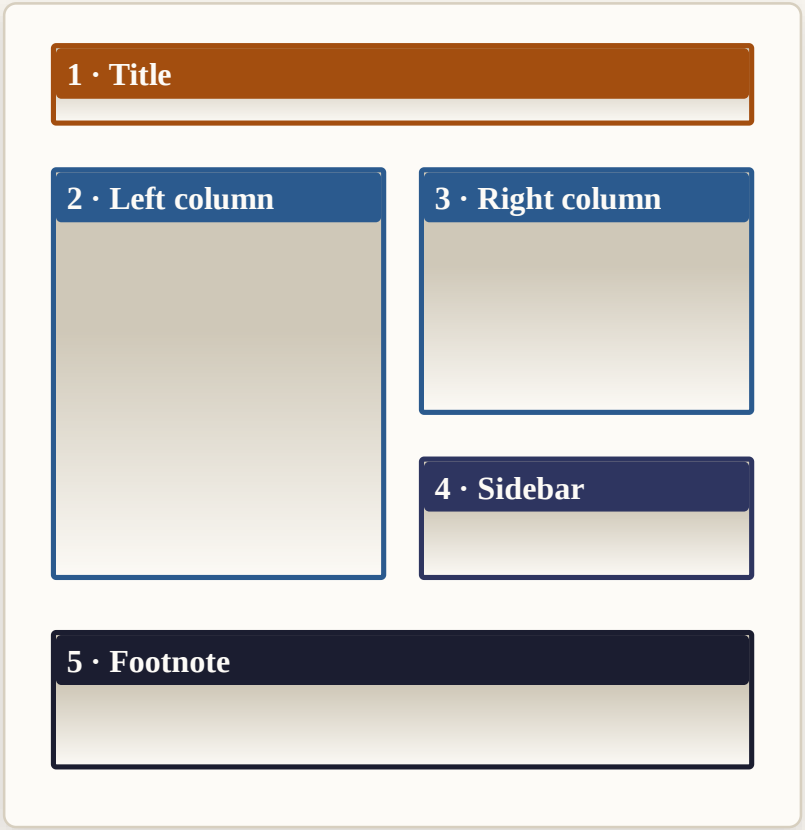
Modern recognisers usually read a **whole line** at once. They see more context and avoid errors from cutting the line into characters.

PITFALL

Character segmentation was the weak point of older OCR: touching letters were split wrongly and every later step inherited the mistake.

READING ORDER

Putting regions in the right sequence



DEFINITION

Reading order is the sequence a human would read the regions in. It is essential for any NLP that follows.

PITFALL

Different languages and documents differ: right to left scripts, vertical Japanese text, forms where labels sit beside values.

PRO TIP

Rules work for simple layouts (columns left to right, top to bottom). Learned reading order models help for magazines, newspapers and complex forms.

TABLES

Table detection and structure recognition

Item	Qty	Price
Paper	2	£4.00
Toner	1	£38.50
Total		£46.50

DEFINITION

Table detection finds where tables are on the page. **Table structure recognition** finds rows, columns, header cells and merged cells.

PITFALL

Tables without ruling lines, cells that span several columns, and tables that continue across pages are the hardest cases.

KEY IDEA

Output a real grid (rows and columns), not just text. Otherwise the numbers lose the meaning that their position gave them.

SPECIAL REGIONS

Other things layout analysis must handle

DEFINITION

Headers and footers: repeated on every page; often excluded from content.

DEFINITION

Stamps and signatures: overlap text; need detection and often separation.

DEFINITION

Checkboxes: detect them and decide ticked or empty.

DEFINITION

Barcodes and QR codes: decode with dedicated readers, not OCR.

DEFINITION

Mathematical formulae: need specialised recognisers.

DEFINITION

Handwritten annotations: may need a separate handwriting model.

PRO TIP

Build an inventory of region types in a sample of your real documents before choosing any tool. Surprises in production usually come from region types nobody listed.

SECTION 4 RECAP

Layout and text detection

REMEMBER

Takeaways

- Find regions before reading them
- Connected components and profiles suit clean print
- Neural detectors handle curved and scene text
- Read lines, not characters, where possible
- Reading order and tables decide downstream meaning

CHECK YOURSELF

Extracted text from a newsletter mixes sentences from neighbouring columns. Recognition or layout problem?

ANSWER

Layout. The characters are read correctly but in the wrong order. Fix column detection and reading order.

SECTION 05

05 Text recognition

How models turn an image of a line into a string of characters: from templates to CNNs, recurrent networks, CTC, attention and transformers.

Template matching **Features** **CRNN** **CTC** **Attention decoders** **Transformers** **Handwriting**

EARLY METHODS

Template matching and hand made features

Template matching

Compare each character image with stored images of every letter in a known font; choose the closest. Works only for fonts it has seen.

Feature extraction

Measure properties: loops, line endings, stroke directions, zoning densities. Feed them to a classifier such as an SVM.

Why they faded

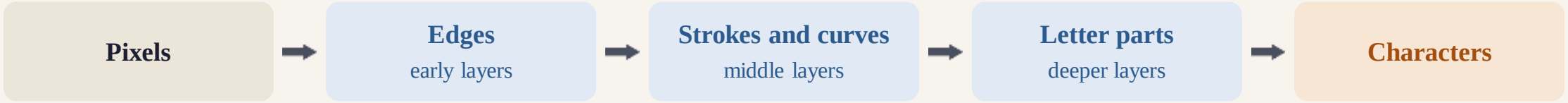
Hand designed features cannot capture the variety of real fonts, handwriting and noise.

ANALOGY

Template matching is like recognising friends only from one passport photo each. Change the lighting or haircut and you fail. Learned features are like knowing a friend from thousands of glimpses.

THE BREAKTHROUGH

Learning features with convolutional networks



DEFINITION

A **convolutional neural network (CNN)** slides small learned filters across the image. Early filters detect edges; deeper ones combine them into strokes, loops and letter shapes.

KEY IDEA

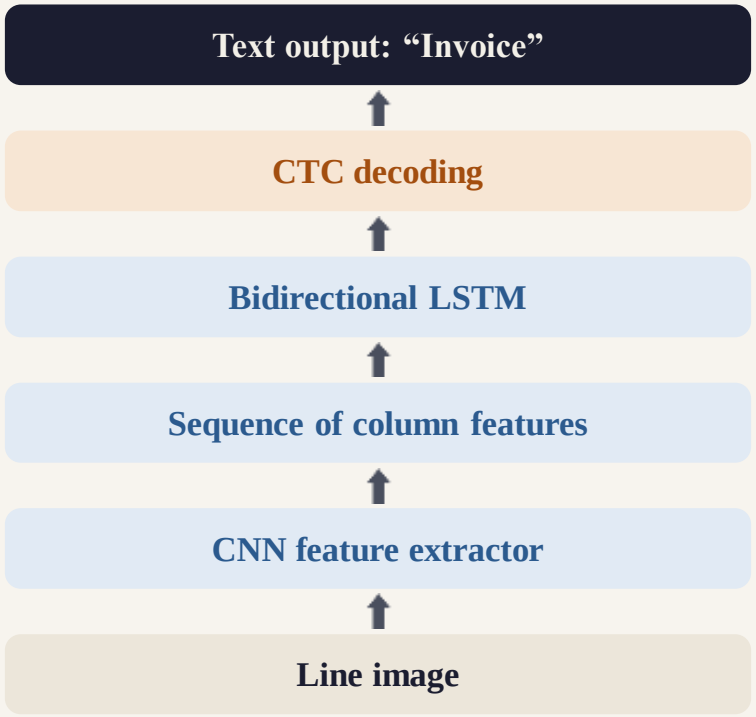
Nobody tells the network what a loop or serif is. It discovers useful patterns from labelled examples.

ANALOGY

Like learning to read as a child: first you notice lines and curves, then letters, then whole words at a glance.

THE WORKHORSE

CRNN: convolutional plus recurrent



DEFINITION

CRNN reads a line image left to right. The CNN turns it into a sequence of feature columns; a recurrent network reads that sequence with context in both directions; CTC turns the result into text.

KEY IDEA

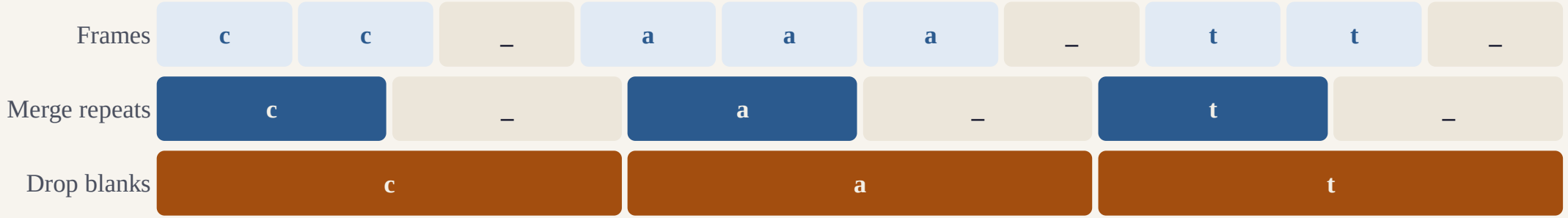
No character segmentation needed. The model learns where letters are while learning what they are.

USE IT WHEN

Printed and handwritten lines, number plates, receipts. Still a strong, efficient baseline.

CTC EXPLAINED

Connectionist temporal classification, visually



DEFINITION

The model predicts a character (or a special **blank**) for every narrow vertical slice of the line. CTC merges repeated predictions and removes blanks to produce the final text.

KEY IDEA

CTC lets you train with just the line image and its text. You never have to mark where each character begins and ends.

WORKED EXAMPLE

Why blanks matter: the word “book” needs a blank between the two o’s, otherwise merging would collapse them into one o and produce “bok”.

Decoding: greedy and beam search

Greedy decoding

Take the most likely symbol for each slice, then merge and drop blanks. Fast and usually good.

Beam search

Keep several promising candidate strings at each step and choose the best overall. Slower, more accurate.

With a language model

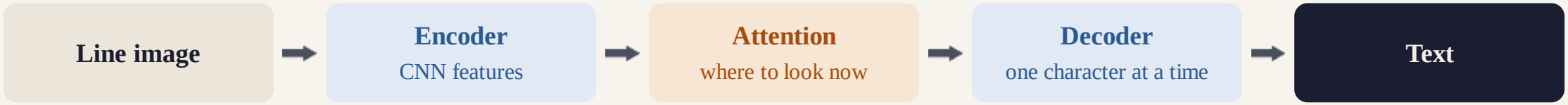
Score candidates by how plausible they are as words in the language. Fixes many look alike errors.

WORKED EXAMPLE

Greedy output: “rn0dern”. Beam search with a word list: “modern”. The visual evidence for rn and m is similar; language knowledge breaks the tie.

ATTENTION

Encoder decoder recognisers with attention



DEFINITION

The decoder writes the text one character at a time. Before each character, **attention** chooses which part of the image to focus on.

ANALOGY

Like reading with your finger under the line: at each step your finger points to the next letter.

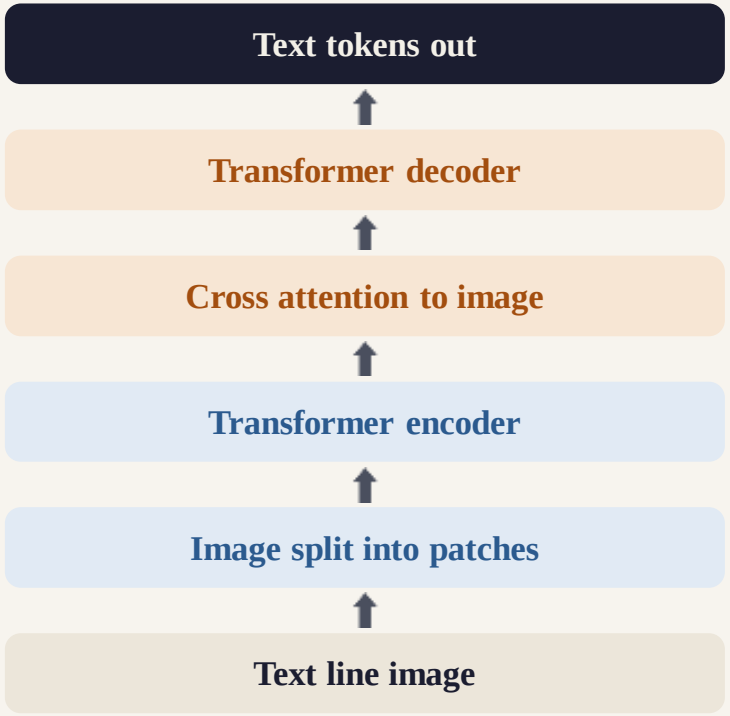
USE IT WHEN

Curved scene text, irregular spacing, text where reading direction is not simply left to right.

PITFALL

Attention decoders can drift, repeating or skipping characters on long lines. CTC is often more stable for long document lines.

Transformer based OCR



DEFINITION

Transformer OCR models (TrOCR is a well known example) split the image into patches, encode them with a vision transformer, and generate text with a language model style decoder.

KEY IDEA

Because the decoder is pretrained on huge amounts of text, it brings strong language knowledge: it knows what words and sentences usually look like.

PITFALL

Larger and slower than CRNN. Strong language priors can also “correct” text that was actually unusual, such as codes and names.

COMPARISON

Recogniser families side by side

Family	Strength	Weakness	Good for
Template and features	Simple, explainable	Brittle with new fonts	Fixed fonts, legacy systems
CRNN with CTC	Fast, stable, small	Weaker language knowledge	Documents, receipts, plates
Attention decoder	Handles irregular text	Can drift on long lines	Scene text
Transformer OCR	Strong accuracy and context	Heavy compute	Handwriting, hard print
Vision language model	Reads whole pages, follows instructions	Costly, can hallucinate	Complex documents, Q and A

HANDWRITING

Handwritten text recognition

WARNING SIGNS

Every writer forms letters differently

WARNING SIGNS

Letters join in cursive

WARNING SIGNS

Slant, size and spacing vary

WARNING SIGNS

Crossings out and insertions

KEY IDEA

Recognise whole lines with CTC or transformer models, trained on large collections of real handwriting from many writers.

PRO TIP

Fine tune on a few hundred lines from your own documents and writers. Gains on a specific archive or form type can be large.

IN THE WILD

Heritage projects transcribe centuries of handwritten records by having experts correct a model’s output, then retraining on the corrections. Each round improves the model.

SCENE TEXT

Reading text in the wild

Signs and shopfronts

Perspective, decorative fonts, reflections.

Product packaging

Curved surfaces, small print, glossy plastic.

KEY IDEA

Scene text systems use strong detectors (often polygon based), rectification to straighten text, and recognisers trained on huge synthetic datasets.

Number plates

Fixed formats, motion blur, night lighting.

Video frames

Low resolution, but many frames to combine.

PRO TIP

In video, read several frames and vote. Agreement across frames is a powerful confidence signal.

TRAINING DATA

Where recognition models learn from

Real labelled data

Images with human transcriptions. Best quality, costly to collect.

Synthetic data

Render text in many fonts, sizes, backgrounds and distortions. Unlimited and free of labelling cost.

Augmentation

Blur, rotate, add noise, change contrast on real images to simulate harder conditions.

PITFALL

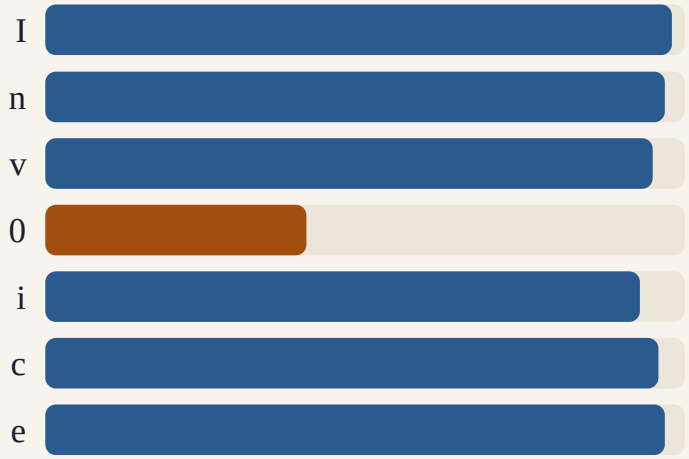
Models trained only on synthetic text can fail on real scans because real noise looks different. Always validate on real documents.

RULE OF THUMB

A common recipe: pretrain on large synthetic data, then fine tune on a smaller set of real, in domain examples.

CONFIDENCE

Knowing when the model is unsure



DEFINITION

Recognisers can output a **confidence** for each character, word or line, based on how strongly the model preferred its answer.

KEY IDEA

Low confidence characters are where to focus checks: send them to correction rules, a second model, or a human reviewer.

PITFALL

Raw confidences are often poorly calibrated. Check how often 90% confident words are actually right before using thresholds.

SECTION 5 RECAP

Text recognition

REMEMBER

Takeaways

- CNNs learn letter shapes from data
- CRNN plus CTC reads lines without segmentation
- Attention and transformers add context and flexibility
- Handwriting and scene text need specialised data
- Use confidence to route uncertain text for checking

CHECK YOURSELF

Why does CTC need a blank symbol?

ANSWER

To separate repeated letters and mark slices with no character. Without blanks, double letters like oo would collapse into one.

06

SECTION 06

Postprocessing and correction

Using dictionaries, language models, formats and business rules to catch and fix recognition errors.

Error types | Dictionaries | Language models | Field rules | Confidence routing | Human review

ERROR TYPES

The four ways OCR gets text wrong

S

Substitution

One character read as another: “tota1” for “total”, “O” for “0”.

D

Deletion

A character is missed: “Invce” for “Invoice”. Common with faint ink.

I

Insertion

An extra character appears, often from noise: “In.voice”.

W

Segmentation

Words merged or split: “invoicetotal” or “in voice”.

KEY IDEA

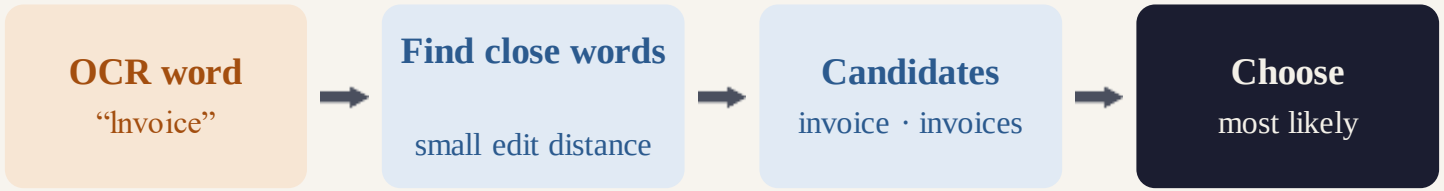
Postprocessing works best when you know which error types dominate. Count them on a labelled sample before choosing fixes.

PRO TIP

Keep a list of the most common confusions in your own data, such as 0 and O or rn and m. Targeted rules for these often beat generic correction.

DICTIONARIES

Lexicon based correction



DEFINITION

Edit distance counts the fewest single character changes (insert, delete, substitute) needed to turn one word into another. “lnvoice” is one change from “invoice”.

PITFALL

Names, codes, postcodes and product numbers are not in any dictionary. Forcing them to the nearest real word creates new errors.

PRO TIP

Use domain dictionaries: your product catalogue, customer names, supplier list, medical vocabulary. They are far more useful than a general English word list.

RULE OF THUMB

Correct only when the OCR word is not a valid entry **and** a candidate is very close **and** confidence was low.

LANGUAGE MODELS

Using context to correct

Please | remit | payment | by | the | due | dale

date?

KEY IDEA

“dale” is a real word, so a dictionary will not flag it. A language model notices that “by the due dale” is very unlikely and “by the due date” is very likely.

DEFINITION

A **language model** estimates how likely a sequence of words is. It can rank OCR candidates by how natural they sound in context.

WORKED EXAMPLE

Modern pipelines pass OCR text with its uncertain spans to a language model and ask for the most plausible correction, with instructions not to change names, numbers or codes.

PITFALL

Large language models may rewrite text to sound fluent, silently changing meaning or inventing words. Restrict them to flagged spans and log every change.

FIELD RULES

Formats and checksums catch impossible values

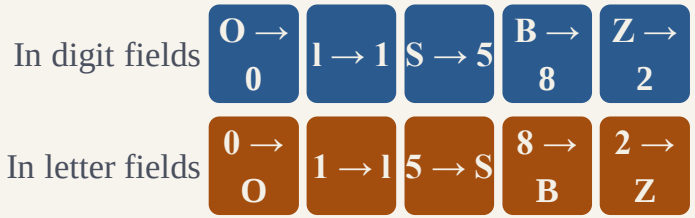
Field	Rule	Example catch
Date	Valid calendar date in expected range	31/02/2026 cannot exist
UK postcode	Matches postcode pattern	SWIA 1AA should be SW1A 1AA
Currency	Digits, one decimal point, two decimals	£1O5.0O becomes £105.00
IBAN and card numbers	Built in check digits	One wrong digit fails the check
Invoice totals	Lines add up to subtotal; tax matches rate	Misread digit breaks the sum
VAT number	Country format and check digits	Extra digit detected

KEY IDEA

Business rules turn silent errors into detected errors. Cross field checks, such as totals that must add up, are among the most powerful OCR validators.

CHARACTER CONFUSIONS

Context specific substitution



DEFINITION

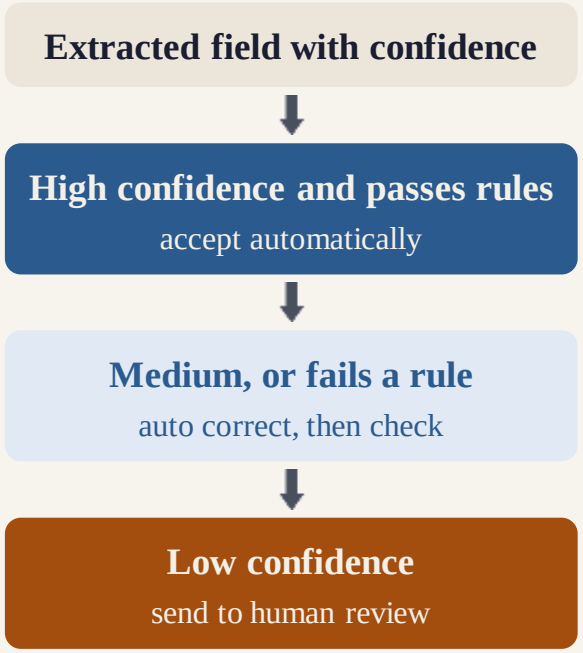
When you know a field should contain only digits (an account number) or only letters (a surname), swap the common look alike characters into the allowed set.

PITFALL

Mixed fields such as postcodes, vehicle registrations and product codes have **position specific** rules: the first two characters may be letters, the next digits. Encode the pattern, not a blanket swap.

ROUTING

Confidence based routing to humans



KEY IDEA

Not every field needs a human, and not every field can skip one. Route by confidence and by business risk.

PRO TIP

Set thresholds per field. A wrong postcode is an annoyance; a wrong bank account number sends money to a stranger.

IN THE WILD

An accounts payable team reviewed every invoice by hand. After routing, people checked only flagged fields, a small share of the total, with no rise in errors reaching payment.

HUMAN IN THE LOOP

Designing review screens

- Show the image crop next to the extracted value
- Highlight low confidence characters
- Pre fill with the best guess
- Keyboard shortcuts for accept and next
- Explain why a field was flagged
- Log every correction

KEY IDEA

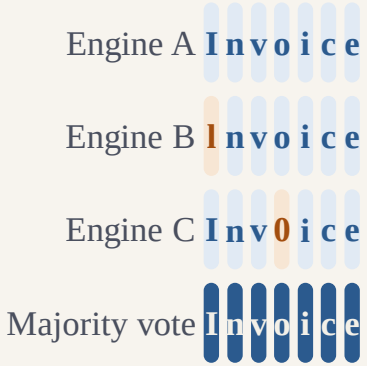
Every human correction is a free training label. Feed corrections back to retrain and to tune thresholds.

PITFALL

If reviewers almost always accept the suggestion, they may stop looking carefully. Insert occasional known test items to check attention.

MERGING ENGINES

Ensembles and voting



DEFINITION

Run two or three different engines (or the same engine on differently preprocessed images), align their outputs and vote character by character.

PRO TIP

Disagreement between engines is an excellent signal for human review, often better than any single engine’s own confidence.

SECTION 6 RECAP

Postprocessing and correction

REMEMBER

Takeaways

- Know your dominant error types
- Domain dictionaries beat general ones
- Language models fix real word errors in context
- Format rules and checksums catch impossible values
- Route by confidence and risk; learn from corrections

CHECK YOURSELF

OCR reads a surname as “Sm1th”. What should a postprocessor do?

ANSWER

In a letters only field, swap 1 to l or i and check against a names list or the customer database. Flag if still uncertain.

SECTION 07

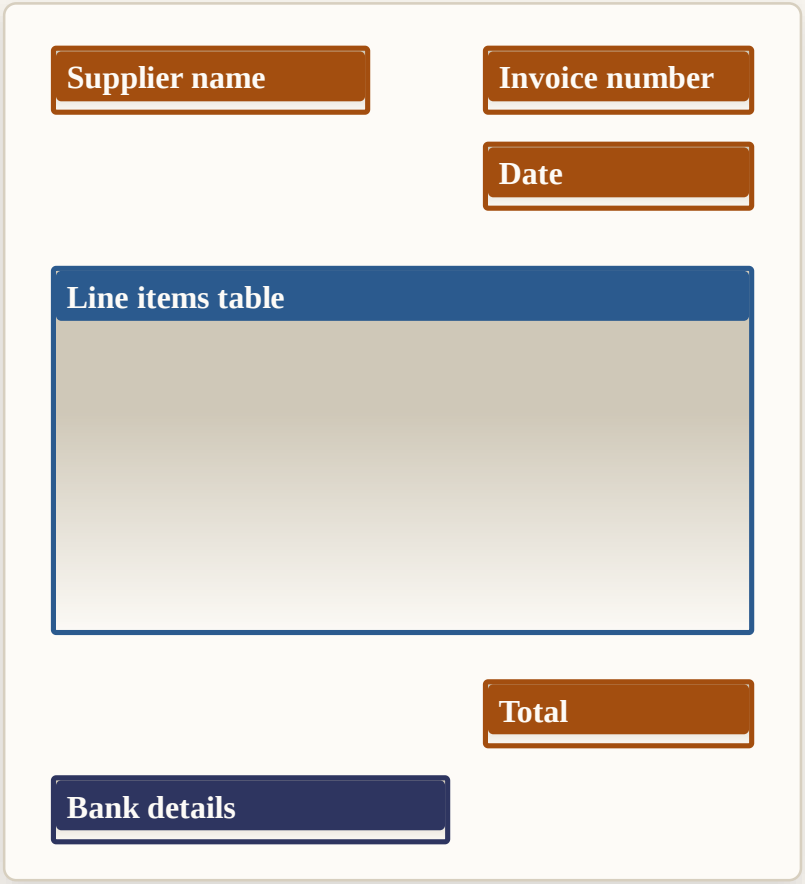
07 Document AI

Beyond plain text: extracting fields, key values, tables and meaning from documents using layout aware and OCR free models.

- Key value extraction
- Templates
- Layout aware models
- OCR free models
- Vision language models
- Classification

THE GOAL

From text to structured data



Field	Value
supplier	Acme Office Ltd
invoice_no	INV 20431
date	14 Oct 2026
total	£1,284.00
lines	6 items

KEY IDEA

Businesses rarely want raw text. They want **fields** in a database, ready to approve, pay, file or analyse.

APPROACH 1

Templates and zonal OCR

DEFINITION

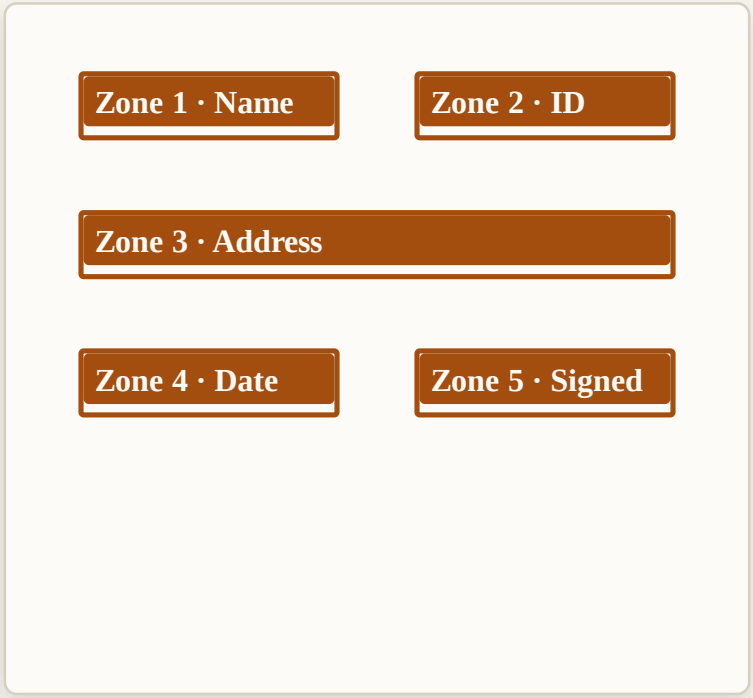
For a fixed form, define zones by position: “the total is always in this rectangle”. Run OCR only inside each zone.

USE IT WHEN

One layout that never changes: a government form, an internal timesheet.

PITFALL

Breaks as soon as the layout shifts, a supplier changes their invoice design, or a scan is offset. Maintaining hundreds of templates is costly.



APPROACH 2

Rules over OCR text

PYTHON · REGULAR EXPRESSION

```
import re
pattern = r"Invoice\s*(No|Number)[:\s]*([\w ]+)"
match = re.search(pattern, text, re.IGNORECASE)
invoice_no = match.group(2).strip() if match else None
```

USE IT WHEN

Moderate layout variation, clear labels, quick wins without training data.

DEFINITION

Search the recognised text for anchor words (“Invoice No”, “Total”) and grab the value next to them using patterns.

PITFALL

Labels vary (“Amount due”, “Balance”, “To pay”), values may sit below rather than beside labels, and OCR errors break patterns.

APPROACH 3

Learning extraction as token labelling

Invoice	No:	INV	20431	Total	due:	£1,284.00
		B NUMBER	*I NUMBER*			*B TOTAL*

DEFINITION

Treat extraction like named entity recognition: label each OCR word as the beginning (B) or inside (I) of a field, or outside any field.

KEY IDEA

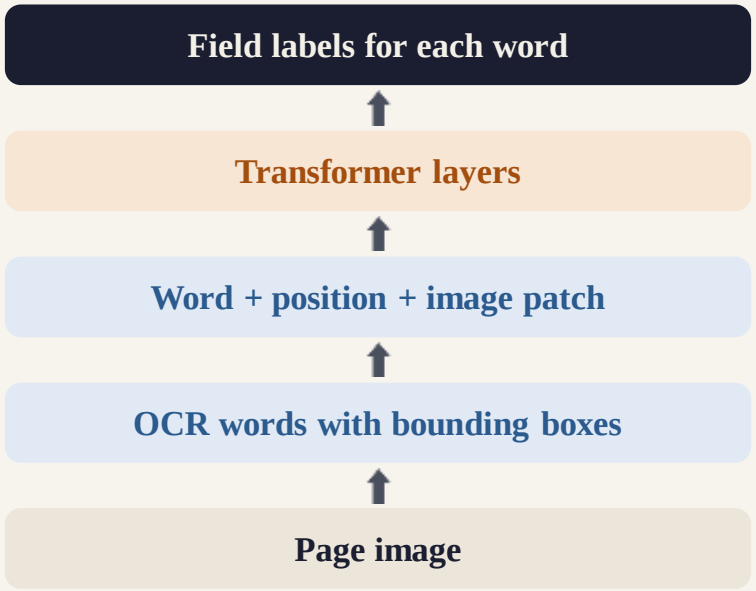
The model learns the many ways a field can appear, instead of you writing a rule for each one.

PRO TIP

This turns document extraction into an NLP task. Part B explains token labelling in depth; here the words come with page positions too.

LAYOUT AWARE MODELS

Reading words and positions together



DEFINITION

Layout aware models (the LayoutLM family is the best known) take each OCR word together with its **box coordinates** and often the image region, so the model knows where on the page every word sits.

ANALOGY

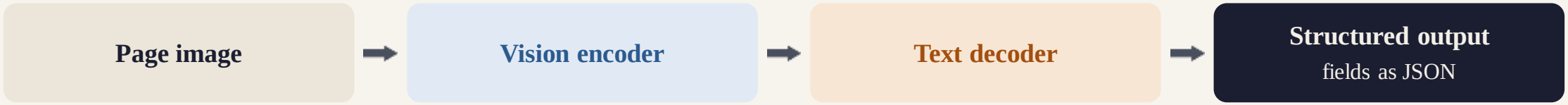
A person finds the total partly by reading “Total” and partly by knowing it sits at the bottom right. These models learn both cues.

USE IT WHEN

Invoices, receipts, forms and contracts with many layouts.

OCR FREE MODELS

Reading documents without a separate OCR step



DEFINITION

OCR free models (Donut is a well known example) go straight from the page image to the desired output, such as a set of fields, without producing intermediate OCR text.

KEY IDEA

No OCR errors to propagate and a simpler pipeline. One model learns reading and understanding together.

PITFALL

Needs more training data and compute; harder to tell whether an error came from reading or understanding; no word boxes for highlighting evidence unless added.

USE IT WHEN

Large volumes of a document type with plenty of labelled examples.

Asking questions of a page

Prompt: What is the invoice total and due date?

Answer: £1,284.00, due 14 November 2026

KEY IDEA

Very fast to prototype: no training, many document types, natural language instructions.

DEFINITION

Vision language models accept an image and a text instruction and answer in text. They can read, interpret layout, and follow instructions such as “return JSON with these fields”.

PITFALL

They can **hallucinate** plausible values that are not on the page, and cost more per page. Validate outputs with rules and require evidence such as the source text.

CHOOSING AN APPROACH

Document AI options compared

Approach	Data needed	Flexibility	Cost to run	Best fit
Templates	None	Very low	Very low	One fixed form
Rules on OCR text	None	Low	Low	Clear labels, few layouts
Token labelling	Hundreds of pages	Medium	Low	Repeated document types
Layout aware	Hundreds to thousands	High	Medium	Many layouts, high volume
OCR free	Thousands	High	Medium	Very high volume, one task
Vision language	Few or none	Very high	High	Variety, low volume, prototypes

RULE OF THUMB

Start with a vision language model to learn what is possible and to create labels, then move high volume document types to cheaper specialised models if costs demand it.

CLASSIFICATION

Document classification and splitting

Upload
one PDF, 40 pages

→

Split
find document boundaries

→

Classify
invoice, ID, letter

→

Route
right extractor

DEFINITION

Document classification decides what type each document is, from text, layout or image features. **Splitting** finds where one document ends and the next begins inside a bundle.

PITFALL

Mis classification sends a document to the wrong extractor, which then confidently extracts nonsense. Monitor classification accuracy separately.

PRO TIP

Visual cues alone (logos, layout) often classify documents well even before OCR. Combine image and text signals for robustness.

EVIDENCE

Always link extracted values to their source

KEY IDEA

Every extracted field should carry its **evidence**: page number, bounding box and the exact OCR text it came from.

WHY IT MATTERS

Reviewers can verify a value in one glance. Auditors can trace decisions. Errors can be diagnosed as reading or understanding failures.



SECTION 7 RECAP

Document AI

REMEMBER

Takeaways

- Businesses need fields, not raw text
- Templates and rules work for stable layouts
- Layout aware models use words plus positions
- OCR free and vision language models read pages directly
- Always keep evidence for each extracted value

CHECK YOURSELF

You receive invoices from 3,000 suppliers with different layouts. Templates or a learned model?

ANSWER

A learned layout aware model or a vision language model. Templates for 3,000 layouts would be unmaintainable.

SECTION 08

08 Evaluating OCR

Measuring OCR quality honestly: character and word error rates, field accuracy, detection metrics and error analysis.

Ground truth **CER** **WER** **Field accuracy** **Detection metrics** **Error analysis**

GROUND TRUTH

You need a reference to measure against

DEFINITION

Ground truth is the correct transcription of each test image, typed and checked by people, together with correct boxes and fields where needed.

PRO TIP

Double key a sample: two people transcribe independently and a third resolves differences. It reveals how ambiguous your documents really are.

- Representative of real traffic, not cherry picked
- Includes the worst realistic images
- Covers every document type and language
- Agreed rules for spaces, punctuation and case
- Kept separate from training data

CHARACTER ERROR RATE

CER: how many characters are wrong



DEFINITION

CER counts the substitutions, deletions and insertions needed to turn the OCR output into the reference, divided by the number of reference characters. Lower is better.

WORKED EXAMPLE

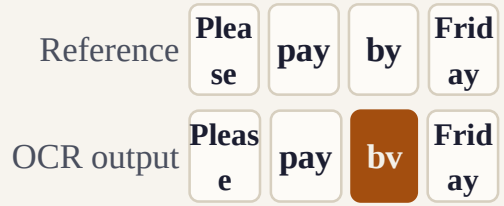
Above: one substitution (I read as l) and one deletion (missing e) out of ten reference characters gives a CER of 20 per cent.

PITFALL

Normalisation choices change CER a lot: case, double spaces, curly versus straight quotes. Fix and document them before comparing systems.

WORD ERROR RATE

WER: how many words are wrong



DEFINITION

WER works like CER but on whole words. One wrong character makes the whole word wrong.

KEY IDEA

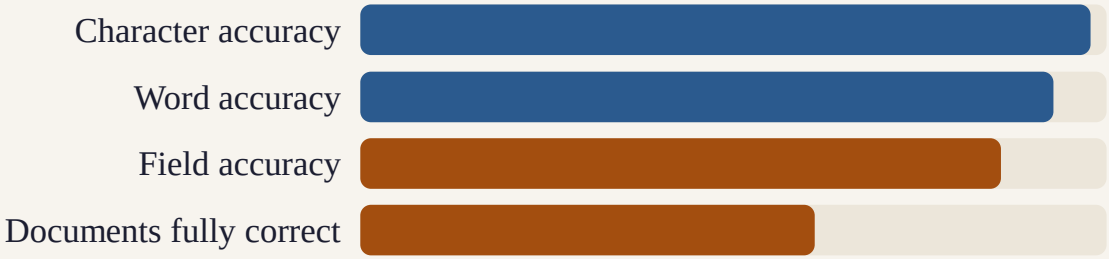
WER is usually several times higher than CER, and is closer to what search and NLP experience. A misspelt word may never match a search query.

WORKED EXAMPLE

Here one character error (y read as v) gives a CER of a few per cent but a WER of 25 per cent: one word in four is wrong.

TASK METRICS

Field level and document level accuracy



KEY IDEA

Small character error rates compound. A document with many fields is only fully correct if every field is right.

RULE OF THUMB

Report the metric the business feels: share of invoices that need no correction, or share of fields entered correctly.

DETECTION METRICS

Measuring how well text is found

DEFINITION

Intersection over union (IoU): overlap between a predicted box and the true box, relative to their combined area.

DEFINITION

A predicted box counts as a hit when IoU passes a threshold such as one half. Then compute **precision** (share of predicted boxes that are real text) and **recall** (share of real text that was found).

PITFALL

A missed line is invisible in CER if you only score lines that were detected. Always score end to end: from page image to final text, including missed regions.

PRO TIP

For reading order, compare the sequence of regions with the reference order, not just the text inside them.

ERROR ANALYSIS

Slice your errors to find the cause

Slice	Question it answers
Document type	Are receipts worse than letters?
Capture source	Are phone photos worse than scans?
Image quality bucket	How fast does accuracy fall with blur or low resolution?
Region type	Are tables, stamps or handwriting the main problem?
Character class	Are digits worse than letters?
Language and script	Is one language far behind?

PRO TIP

Build a confusion table of characters (which characters are read as which). It points directly to fixes such as field rules or targeted training data.

BENCHMARKS

Public datasets and their limits

Printed documents

Collections of scanned pages and receipts used to compare engines and layout tools.

Scene text

Competition datasets of photos with signs and labels, often from the ICDAR series.

Handwriting

Datasets of handwritten lines and pages from many writers, such as the IAM collection.

Forms and receipts

Datasets labelled for key value extraction such as FUNSD, SROIE and CORD style collections.

PITFALL

Benchmark scores rarely predict performance on your documents. Always build a small, honest test set from your own traffic.

SECTION 8 RECAP

Evaluating OCR

REMEMBER

Takeaways

- Build checked, representative ground truth
- CER for characters, WER for words
- Field and document accuracy match business value
- Score detection and reading order end to end
- Slice errors to find causes

CHECK YOURSELF

CER is 2% but a quarter of invoices need correction. Contradiction?

ANSWER

No. Errors spread across many fields per invoice, so even a low CER leaves many documents with at least one wrong field.

SECTION 09

09 OCR tools and pipelines

The main open source engines and cloud services, and how to assemble them into a robust production pipeline.

Tesseract | PaddleOCR | EasyOCR | TrOCR | Cloud APIs | Pipeline design

LANDSCAPE

The OCR tool landscape

Tool	Type	Strength	Consider when
Tesseract	Open source engine	Mature, many languages, runs anywhere	Clean printed documents, offline use
PaddleOCR	Open source toolkit	Strong detection and recognition models	Varied layouts, multilingual, tables
EasyOCR	Open source library	Simple to use, many scripts	Quick prototypes, scene text
docTR and similar	Open source deep learning	End to end detection plus recognition	Python pipelines on GPUs
TrOCR and similar	Transformer models	Handwriting and hard print	Line level accuracy matters most
Cloud document APIs	Managed services	Layout, tables, forms, handwriting at scale	No ML team, fast time to value

☆ REMEMBER

Tools evolve quickly. Check current documentation, licences and data handling terms before choosing.

TESSERACT

Using Tesseract well

PYTHON · PYTESSERACT

```
import pytesseract
from PIL import Image

img = Image.open("page.png")
text = pytesseract.image_to_string(img, lang="eng")

# word boxes and confidences
data = pytesseract.image_to_data(
    img, output_type=pytesseract.Output.DICT)
words = [(w, c) for w, c in
         zip(data["text"], data["conf"]) if w.strip()]
```

PRO TIP

- Feed it clean, deskewed, high contrast images
- Set the page segmentation mode to match your layout
- Install the right language packs
- Use word confidences to route review

PITFALL

Tesseract struggles with photos, handwriting and complex tables without help. Pair it with good preprocessing and layout detection.

DEEP LEARNING TOOLKITS

Detection plus recognition in a few lines

PYTHON · PADDLEOCR (API VARIES BY VERSION)

```
from paddleocr import PaddleOCR

ocr = PaddleOCR(lang="en")
result = ocr.ocr("receipt.jpg")
for box, (text, score) in result[0]:
    print(text, round(score, 2))
```

KEY IDEA

Toolkits like PaddleOCR return each text region’s polygon, its text, and a confidence score, ready for layout reconstruction and routing.

PRO TIP

Pin library versions. OCR toolkit interfaces change between releases and silently break pipelines.

CLOUD SERVICES

Managed document AI services

General OCR

Text, lines and words with boxes from images and PDFs.

Forms and tables

Key value pairs and table structure.

USE IT WHEN

You need many document types quickly, have limited ML staff, and can send data to a provider.

Prebuilt models

Invoices, receipts, IDs and other common documents.

Custom models

Train on your own labelled documents.

PITFALL

Check data residency, retention and privacy terms, especially for health, financial and identity documents under UK GDPR.

BUILD OR BUY

A decision guide

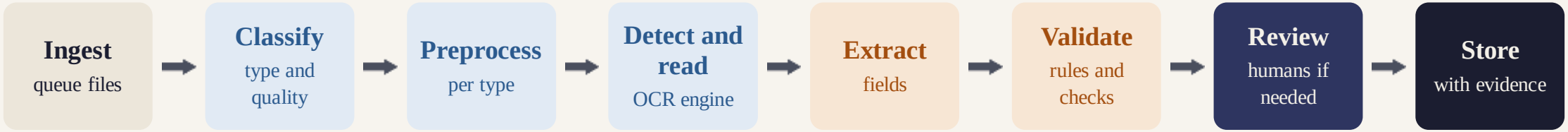
Factor	Favours open source	Favours cloud service
Data sensitivity	Must stay on premises	Provider terms acceptable
Volume	Very high, cost per page matters	Low to moderate
Team skills	ML engineers available	Small or no ML team
Customisation	Unusual documents or scripts	Common document types
Time to value	Months acceptable	Weeks needed
Control	Need to inspect and tune everything	Managed updates acceptable

PRO TIP

Benchmark two or three options on the same 200 of your own documents, using field accuracy and cost per page. Decide with evidence, not marketing claims.

PIPELINE DESIGN

A production OCR pipeline



KEY IDEA

Make each stage a separate, observable step with its own logs and metrics. When accuracy drops, you can see exactly where.

PRO TIP

Save intermediate outputs (preprocessed image, boxes, raw text) for a sample of documents. They are invaluable for debugging.

PIPELINE DESIGN

Engineering details that matter

DEFINITION

Idempotency: reprocessing the same file gives the same result and no duplicates.

DEFINITION

Retries and dead letter queues: failed pages are retried, then parked for inspection.

DEFINITION

Versioning: record model, rules and preprocessing versions with every result.

DEFINITION

Throughput: batch pages on GPUs; scale workers with queue length.

DEFINITION

Monitoring: confidence distributions, rule failures, review rates per document type.

DEFINITION

Security: encrypt files at rest and in transit; limit who can see raw documents.

SECTION 9 RECAP

OCR tools and pipelines

REMEMBER

Takeaways

- Tesseract for clean print; deep toolkits for variety
- Cloud services trade control for speed
- Benchmark on your own documents
- Design observable, versioned pipeline stages
- Plan for privacy and security from day one

CHECK YOURSELF

Health records must never leave your servers. Which option fits?

ANSWER

Open source engines deployed on premises or in your own controlled environment, or a provider deployment that meets your data residency requirements.

PART B

Part B · Natural Language Processing

Nine sections on how machines work with language: foundations, tokenisation, representations, classical tasks, sequence models, transformers, large language models, applications and evaluation.

SECTION 10

10 NLP foundations

What natural language processing is, the layers of language, why ambiguity makes it hard, and how the field evolved.

Definition | Levels of language | Ambiguity | History | Task map

DEFINITION

Natural language processing

DEFINITION

Definition

NLP is the field of computer science and AI that enables machines to process, understand and generate human language, in text or speech.

IN PLAIN WORDS

In plain words

Teaching computers to do useful things with words: find, sort, extract, summarise, translate, answer and write.

Text in
emails, documents, OCR output

→

Understand
classify, extract, link

→

Reason
compare, answer, decide

→

Generate
summaries, replies, translations

→

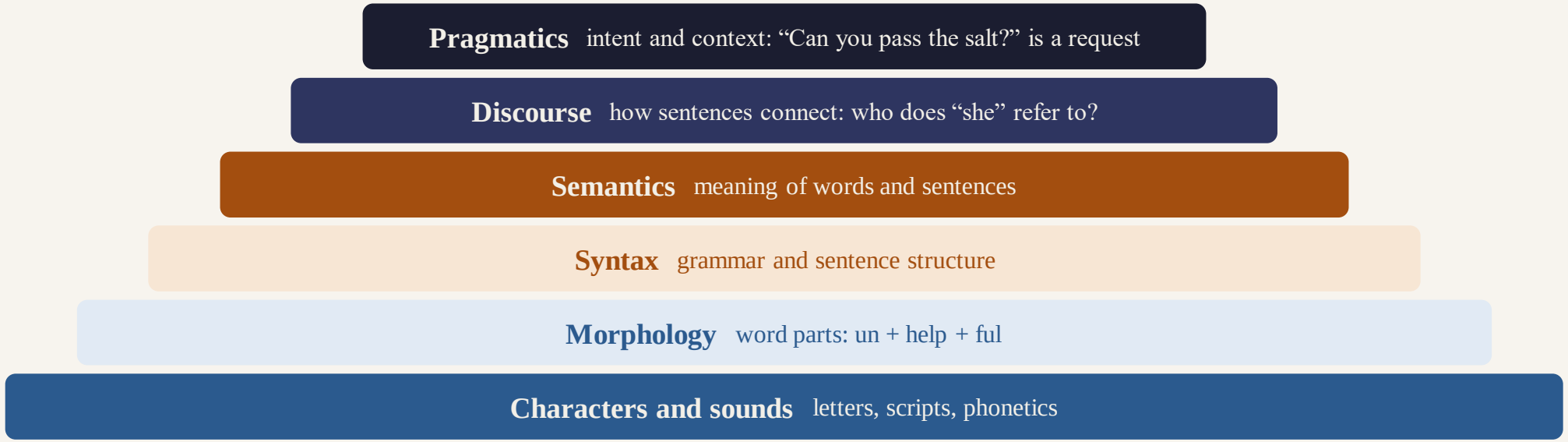
Value
faster, cheaper, better decisions

ANALOGY

If OCR gives a computer eyes for print, NLP gives it a basic grasp of language: grammar, vocabulary and some sense of meaning.

LAYERS OF LANGUAGE

Language has many levels



KEY IDEA

Classical NLP built separate tools for each level. Modern neural models learn all levels together from huge amounts of text, though the levels remain useful for diagnosing errors.

AMBIGUITY

Why language is hard for machines

WORKED EXAMPLE

Lexical: “bank” can mean a river edge or a financial institution.

WORKED EXAMPLE

Syntactic: “I saw the man with the telescope.” Who has the telescope?

WORKED EXAMPLE

Referential: “The trophy did not fit in the case because it was too big.” What was too big?

WORKED EXAMPLE

Pragmatic: “Nice weather!” said during a storm is sarcasm.

WORKED EXAMPLE

Scope: “Every student read a book.” The same book or different ones?

WORKED EXAMPLE

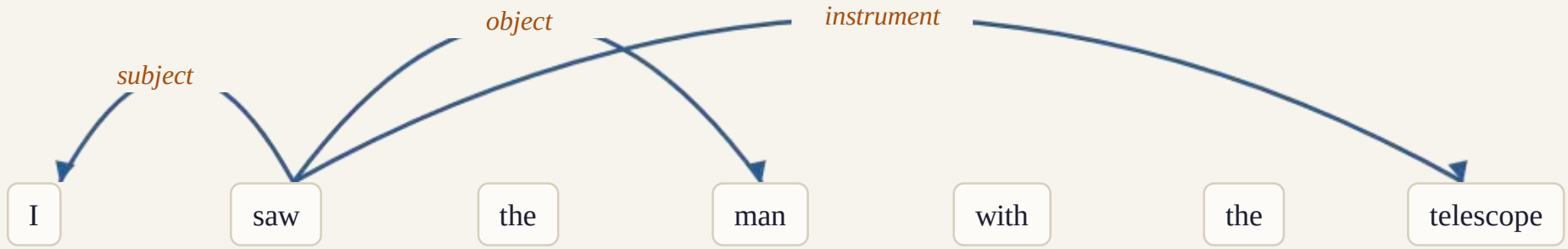
Ellipsis: “I will if you will.” Will what?

KEY IDEA

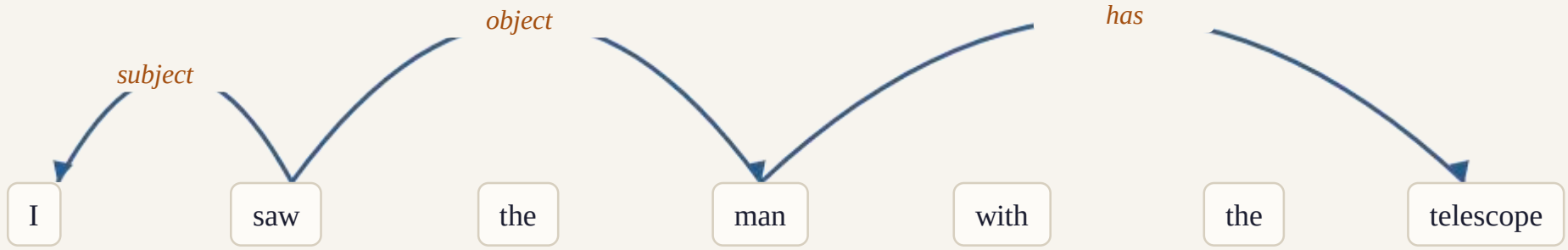
Humans resolve ambiguity effortlessly with world knowledge and context. Much of NLP history is the search for ways to give machines that context.

AMBIGUITY

One sentence, two structures



Reading 1: I used the telescope to see the man.



Reading 2: the man I saw was holding a telescope.

HISTORY

Seventy years of NLP

1950s to 1980s

Rules

Hand written grammars and dictionaries; early machine translation; ELIZA style chatbots.

1990s to 2000s

Statistics

Probabilities learned from text corpora: n gram models, hidden Markov models, statistical translation.

2013

Word embeddings

Word2vec and GloVe represent words as vectors that capture meaning.

2014 to 2017

Neural sequences

RNNs, LSTMs, sequence to sequence and attention.

2017 onwards

Transformers and LLMs

Attention is all you need; BERT, GPT and large language models.

KEY IDEA

Each era kept the useful parts of the previous one. Tokenisation, evaluation habits and many classical tasks remain essential today.

PRO TIP

Classical methods are still the right answer for many small, well defined problems: they are cheap, fast and explainable.

TASK MAP

The main families of NLP tasks

Classification

Assign labels to text: spam, sentiment, topic, intent, urgency.

Extraction

Pull out entities, relations, dates, amounts, clauses.

Generation

Write summaries, replies, translations, reports.

Matching and retrieval

Find similar documents, answer search queries, deduplicate.

Question answering

Answer questions from a document or a knowledge base.

Structure prediction

Tag parts of speech, parse grammar, segment sentences.

RULES VERSUS LEARNING

Three ways to build an NLP system

Approach	How it works	Strength	Weakness
Rules	Experts write patterns and dictionaries	Precise, explainable, no data needed	Brittle; costly to maintain
Classical ML	Features plus a trained classifier	Cheap, fast, decent accuracy	Needs feature engineering and labels
Deep learning	Models learn features from data	High accuracy, handles nuance	Needs data and compute
LLM prompting	Instruct a pretrained model	Little or no labelled data	Cost, latency, hallucination risk

RULE OF THUMB

Choose the simplest approach that meets the accuracy, cost and explainability your use case needs. A regular expression that is right 99.9% of the time beats any model for a fixed code format.

DATA

Corpora: the fuel of NLP

DEFINITION

A **corpus** (plural corpora) is a collection of text used to train or evaluate NLP systems: emails, articles, transcripts, OCR output, web pages.

KEY IDEA

Models learn the language of their corpus. A model trained on news articles will stumble on medical notes or text messages.

- Is it representative of real inputs?
- Is it labelled consistently?
- Is personal data handled lawfully?
- Is it free of duplicates and leakage?
- Does it cover all languages and dialects you serve?

SECTION 10 RECAP

NLP foundations

REMEMBER

Takeaways

- NLP processes, understands and generates language
- Language has layers from characters to pragmatics
- Ambiguity is the central difficulty
- The field moved from rules to statistics to neural models
- Simple methods still win for narrow problems

CHECK YOURSELF

"The chicken is ready to eat." What kind of ambiguity is this?

ANSWER

Syntactic and semantic: the chicken may be the food or the one about to eat. Context decides.

11

SECTION 11

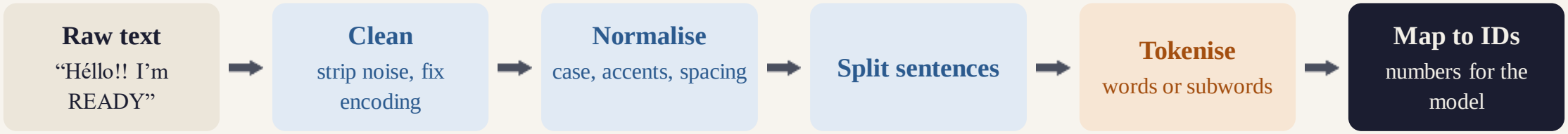
Text preprocessing and tokenisation

Turning raw strings into clean, consistent units that models can count, compare and learn from.

Normalisation · Unicode · Sentences · Word tokens · Subwords · Stemming and lemmas · Stopwords

THE PIPELINE

From raw text to model input



KEY IDEA

Every model works with **numbers**, not letters. Preprocessing and tokenisation decide exactly which pieces of text become which numbers.

PITFALL

Preprocessing at training time and at prediction time must be identical. A tiny difference, such as lowercasing in only one of them, silently ruins accuracy.

CLEANING

Typical cleaning steps

Step	Example before	Example after
Remove markup	Hello	Hello
Fix encoding errors	CafÃ©	Café
Collapse whitespace	too many spaces	too many spaces
Normalise quotes	“smart” and “straight”	one consistent style
Expand or keep contractions	don’t	do not, or keep as is
Mask personal data	Call 07700 900123	Call [PHONE]

PRO TIP

For OCR output, also join words broken across line ends, remove repeated headers and footers, and keep page numbers as metadata rather than text.

NORMALISATION

Case, accents and Unicode

DEFINITION

Unicode gives every character in every script a number. Some characters can be written in more than one way, such as an accented letter as one symbol or as a letter plus an accent mark.

DEFINITION

Unicode normalisation (NFC, NFKC and friends) converts these variants into one standard form so identical looking text compares as equal.

Choice	When it helps	When it hurts
Lowercase	Search, small data	Names, acronyms: US vs us
Strip accents	Noisy user input	Languages where accents change meaning
NFKC	Full width digits, ligatures	When exact form matters

PITFALL

Modern pretrained models come with their own tokenisers that expect raw text with case. Lowercasing before them removes information they were trained to use.

SENTENCES

Sentence segmentation is not trivial

Dr.	Smith	paid	£3.50	at	9 a.m.	Then	left.
not an end			*not an end*		*maybe an end*		*end*

KEY IDEA

Full stops also appear in abbreviations, decimals, times, email addresses and initials. Good sentence splitters combine rules with learned models.

PITFALL

OCR text often has no reliable punctuation or line breaks in the wrong places. Sentence splitting on OCR output needs extra care, using layout cues such as paragraph gaps.

WORD TOKENISATION

Splitting text into words

Input: “Don’t over think it, it’s £10.”

Whitespace	**Don’t**		**over**		**think**		**it,**		**it’s**		**£10.**
Rule based	**Do**	**n’t**	**over**	**think**	**it**	**,**	**it**	**’s**	**£**	**10**	**.**

DEFINITION

A **token** is a unit of text a model treats as one item. Rule based tokenisers separate punctuation, contractions and symbols.

PITFALL

Languages such as Chinese, Japanese and Thai do not separate words with spaces, so word tokenisation needs dictionaries or learned models.

THE VOCABULARY PROBLEM

Why whole words are not enough

WARNING SIGNS

- Vocabularies of whole words grow into millions
- New words appear daily: names, slang, products
- Rare words get too few examples to learn
- OCR errors create endless fake words

DEFINITION

An **out of vocabulary** word is one the model has never seen. Word level models usually replace it with a single unknown token, losing all its information.

KEY IDEA

Subword tokenisation solves this: common words stay whole, rare words split into familiar pieces. Every word can be represented.

tokenisation

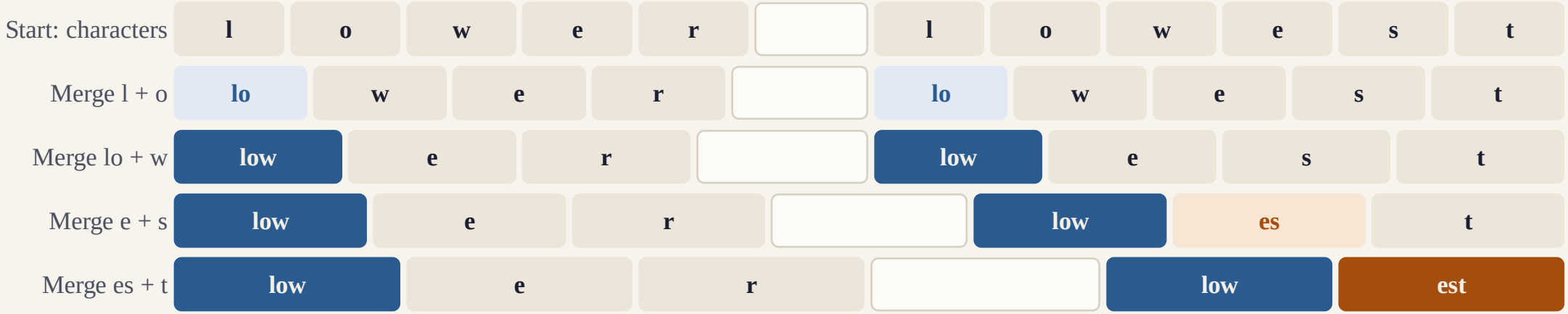
token | isation

unbelievably

un | believ | ably

SUBWORDS

Byte pair encoding, step by step



DEFINITION

BPE starts from single characters and repeatedly merges the most frequent neighbouring pair into a new token, until the vocabulary reaches a chosen size.

KEY IDEA

Frequent words become single tokens; rare words are built from a few pieces. Vocabulary sizes of tens of thousands are typical.

SUBWORDS

WordPiece, Unigram and byte level tokenisers

Method	Idea in plain words	Used by (examples)
BPE	Merge the most frequent pairs	GPT style models
Byte level BPE	Work on raw bytes so any text, emoji or script fits	Many modern LLMs
WordPiece	Merge pairs that most improve how well the vocabulary explains the data	BERT style models
Unigram (SentencePiece)	Start big and prune pieces that matter least	T5 and many multilingual models

KEY IDEA

Always use the tokeniser that was trained with your pretrained model. Mixing tokenisers is like using the wrong dictionary.

PRO TIP

Token counts drive LLM cost and context limits. Non English text and OCR noise often need many more tokens per word.

TOKENS AND COST

How OCR noise inflates token counts

Invoice | total | due

Clean text: 3 tokens

ln | vo | 1 | ce | tot | a1 | d | ue

Noisy OCR: many more tokens

KEY IDEA

Misread words split into strange fragments. Models see unfamiliar pieces, understand less, and you pay for more tokens.

PRO TIP

Correct OCR errors before NLP where you can. Cleaner text improves both accuracy and cost.

WORD FORMS

Stemming versus lemmatisation

Word	Stem (chop endings)	Lemma (dictionary form)
running	run	run
ran	ran	run
studies	studi	study
better	better	good
organisation	organ	organisation

DEFINITION

Stemming cuts word endings with simple rules. Fast but crude: it can produce non words.

DEFINITION

Lemmatisation uses vocabulary and grammar to find the true base form.

USE IT WHEN

Keyword search, topic models and small classical pipelines. Rarely needed with transformers.

STOPWORDS

Removing very common words: when and when not

the | payment | was | not | received

DEFINITION

Stopwords are very frequent words such as the, a, is, of. Classical pipelines often removed them to shrink data.

PITFALL

Standard stopword lists often include “not” and “no”. Removing them flips meaning: “payment was not received” becomes “payment received”.

RULE OF THUMB

Remove stopwords only for keyword style tasks, and check the list. Never remove them for sentiment, extraction or transformer models.

SECTION 11 RECAP

Text preprocessing and tokenisation

REMEMBER

Takeaways

- Same preprocessing in training and production
- Normalise Unicode; think before lowercasing
- Sentence splitting is harder than it looks
- Subword tokenisers handle any word
- Stopword removal can flip meaning

CHECK YOURSELF

Why do subword tokenisers handle OCR errors better than word level vocabularies?

ANSWER

A misread word is not lost as an unknown token; it splits into smaller familiar pieces that still carry some information.

SECTION 12

12

Representing text

How words and documents become numbers: counts, weights, n grams, embeddings and contextual vectors.

One hot · Bag of words · TF IDF · N grams · Word embeddings · Contextual embeddings · Sentence embeddings

THE QUESTION

How do you turn words into numbers?

KEY IDEA

Models do arithmetic. To use text, we must represent each word, sentence or document as a list of numbers, called a **vector**, that keeps as much of the meaning as possible.

ANALOGY

Like describing a city with coordinates. Once every city has latitude and longitude, a computer can measure which cities are near each other.

One hot identity only → **Counts** how often → **TF IDF** how distinctive → **Embeddings** what it means → **Contextual** meaning in this sentence

ONE HOT VECTORS

The simplest representation

cat	0.0	0.0	1.0	0.0	0.0	0.0
dog	0.0	0.0	0.0	1.0	0.0	0.0
invoice	0.0	0.0	0.0	0.0	0.0	1.0

DEFINITION

Each word gets a vector as long as the vocabulary, with a single 1 at its own position and 0 everywhere else.

PITFALL

Every word is equally different from every other. “cat” is no closer to “dog” than to “invoice”. The vectors also become enormous for real vocabularies.

BAG OF WORDS

Counting words in documents

	invoice	paid	late	total	refund
Doc 1	2.0	1.0	0.0	1.0	0.0
Doc 2	0.0	0.0	2.0	1.0	1.0
Doc 3	1.0	0.0	0.0	0.0	2.0

DEFINITION

Represent each document as counts of each vocabulary word. Word order is ignored, like emptying words into a bag.

USE IT WHEN

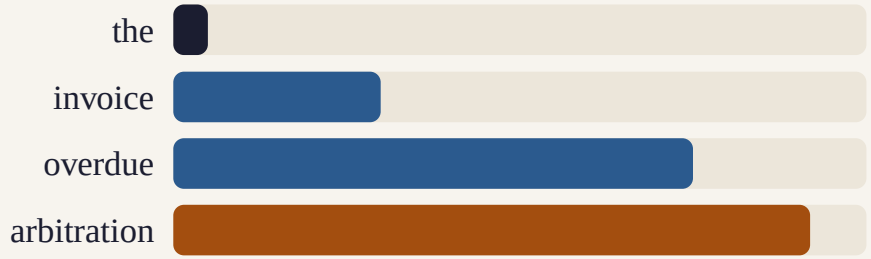
Spam filters, topic routing, baselines. Often surprisingly strong with simple classifiers.

PITFALL

“Dog bites man” and “man bites dog” look identical.

TF IDF

Weighting words by how distinctive they are



DEFINITION

TF IDF gives a word a high weight when it appears often **in this document** (term frequency) but rarely **across all documents** (inverse document frequency).

KEY IDEA

Common words everywhere get low weight; distinctive words that characterise this document get high weight.

USE IT WHEN

Search ranking, document similarity, keyword extraction, fast classification baselines.

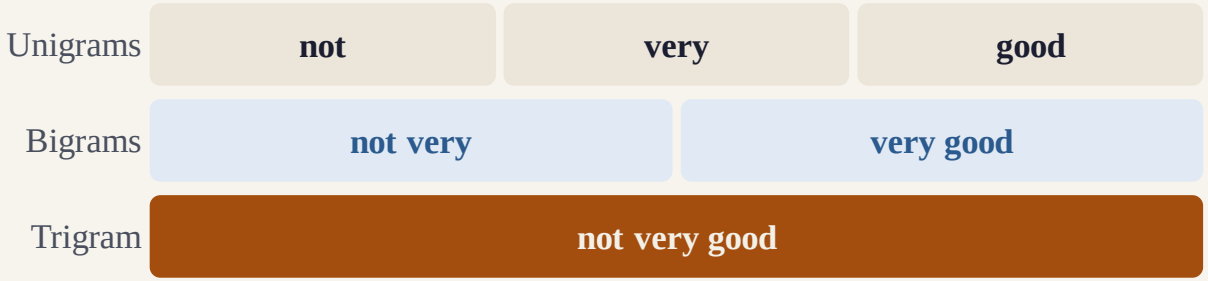
PRO TIP

TF IDF plus a linear classifier is one of the strongest cheap baselines in NLP. Always try it before a large model.

N GRAMS

Capturing a little word order

not very good



DEFINITION

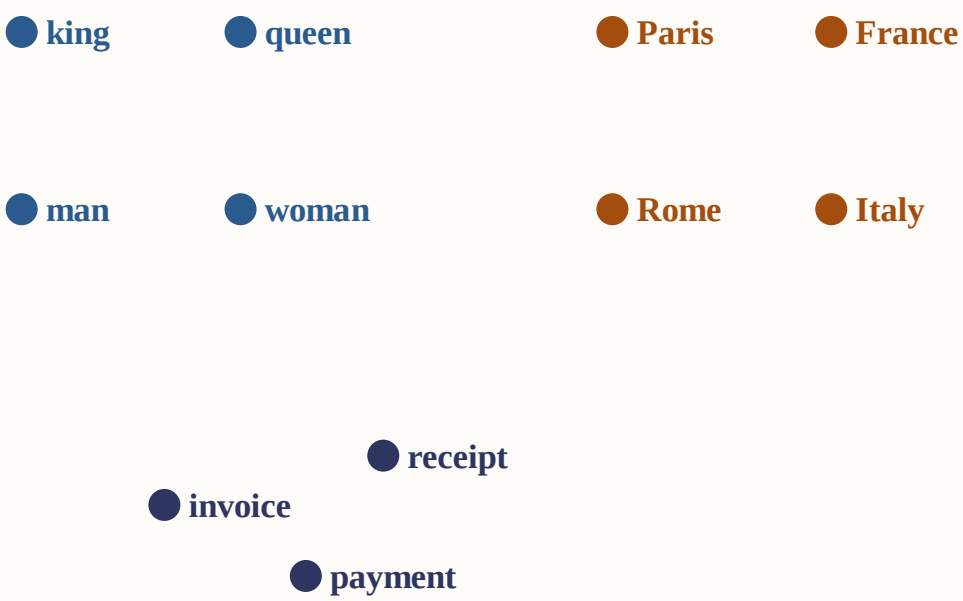
An **n gram** is a run of n consecutive tokens. Bigrams and trigrams preserve short phrases such as “not good” or “New York”.

PITFALL

The number of possible n grams explodes, making vectors huge and sparse. Character n grams help with OCR noise and spelling variation.

WORD EMBEDDINGS

Words as points in meaning space



DEFINITION

An **embedding** is a dense vector of a few hundred numbers learned so that words used in similar contexts get similar vectors.

KEY IDEA

Similar words cluster together, and directions can carry meaning: the step from man to woman resembles the step from king to queen.

ANALOGY

“You shall know a word by the company it keeps.” Words that keep similar company end up as neighbours.

How word2vec learns

customer | paid | the | invoice | on | time

Context words around the centre word “invoice”

Skip gram
predict context from the word

→

CBOW
predict the word from context

📖 **DEFINITION**

Word2vec trains a tiny network on a simple game: guess nearby words. The weights learned to play the game become the word vectors.

📖 **DEFINITION**

GloVe learns vectors from global counts of which words appear together. **fastText** adds character pieces, so it can build vectors for misspelt or new words.

Strengths and limits of static embeddings

REALITY

Capture similarity and some analogies; tiny and fast; great features for small models.

PITFALL

One vector per word, whatever the context: “bank” has the same vector by a river and in finance.

PITFALL

They absorb biases from training text, such as stereotyped associations between occupations and gender.

PRO TIP

fastText style subword embeddings are a good fit for noisy OCR text because misspelt words still share pieces with correct ones.

CONTEXTUAL EMBEDDINGS

The same word, different meanings



DEFINITION

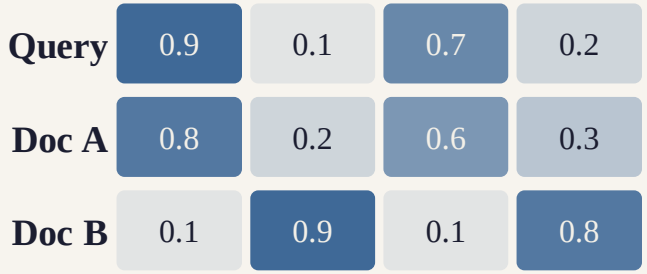
Contextual embeddings give each word a different vector depending on the sentence it appears in. They come from models such as ELMo, BERT and GPT style transformers.

KEY IDEA

“We sat on the bank” and “The bank approved the loan” now place “bank” in different regions of meaning space.

SENTENCE EMBEDDINGS

One vector for a whole sentence or document



Doc A is close to the query; Doc B is far away.

DEFINITION

Sentence embedding models map whole sentences or passages to vectors so that texts with similar meaning are close, even when they use different words.

USE IT WHEN

Semantic search, deduplication, clustering, retrieval augmented generation.

DEFINITION

Cosine similarity measures how closely two vectors point in the same direction. It is the usual way to compare embeddings.

CHOOSING A REPRESENTATION

Representations compared

Representation	Captures	Cost	Use when
Bag of words	Which words occur	Very low	Baselines, simple routing
TF IDF with n grams	Distinctive words and phrases	Low	Search, small data classification
Static embeddings	General word meaning	Low	Small models, noisy text with subwords
Contextual embeddings	Meaning in context	Medium to high	Most modern NLP
Sentence embeddings	Meaning of whole passages	Medium	Semantic search and retrieval

SECTION 12 RECAP

Representing text

REMEMBER

Takeaways

- Models need vectors, not words
- Bag of words and TF IDF are strong cheap baselines
- N grams keep short phrases
- Embeddings place similar meanings close together
- Contextual vectors resolve ambiguity

CHECK YOURSELF

Why does TF IDF give “the” a near zero weight?

ANSWER

It appears in almost every document, so it does not distinguish any document from the others.

SECTION 13

13 Classical NLP tasks

The core analysis tasks every practitioner should know: tagging, entities, parsing, coreference, sentiment, topics and relations.

Part of speech **Named entities** **BIO tagging** **Parsing** **Coreference** **Sentiment** **Topics** **Relations**

PART OF SPEECH TAGGING

Labelling the grammatical role of each word

The	supplier	quickly	sent	a	new	invoice
determiner	*noun*	*adverb*	*verb*	*determiner*	*adjective*	*noun*

DEFINITION

POS tagging assigns each word a grammatical category: noun, verb, adjective, adverb, preposition and so on.

WHY IT MATTERS

It helps parsing, lemmatisation, information extraction and disambiguation: “book” as a noun versus “book a room” as a verb.

WORKED EXAMPLE

“Time flies like an arrow” shows why it is hard: flies can be a verb or a plural noun, and like a verb or a preposition.

NAMED ENTITY RECOGNITION

Finding names of things in text

Anthony	joined	Phexara	in	Nottingham	on	2 June 2026	for	£50,000
PERSON		*ORG*		*LOCATION*		*DATE*		*MONEY*

DEFINITION

NER finds spans of text that refer to entities and labels their type: people, organisations, places, dates, amounts, products, medical terms and more.

KEY IDEA

NER is the backbone of extraction from contracts, invoices, medical notes and news. It is often the first NLP step after OCR.

NER LABELS

The BIO scheme

Bank	of	England	raised	rates	in	March
B ORG	*I ORG*	*I ORG*	*O*	*O*	*O*	*B DATE*

DEFINITION

B marks the beginning of an entity, **I** marks a word inside it, **O** marks words outside any entity. This turns span finding into labelling one token at a time.

PITFALL

With subword tokenisers, one word may become several tokens. Decide whether labels go on the first piece only, and convert back to whole words consistently.

PRO TIP

Evaluate NER at the **entity** level: an entity counts as correct only if its exact span and type are both right.

NER APPROACHES

From rules to transformers

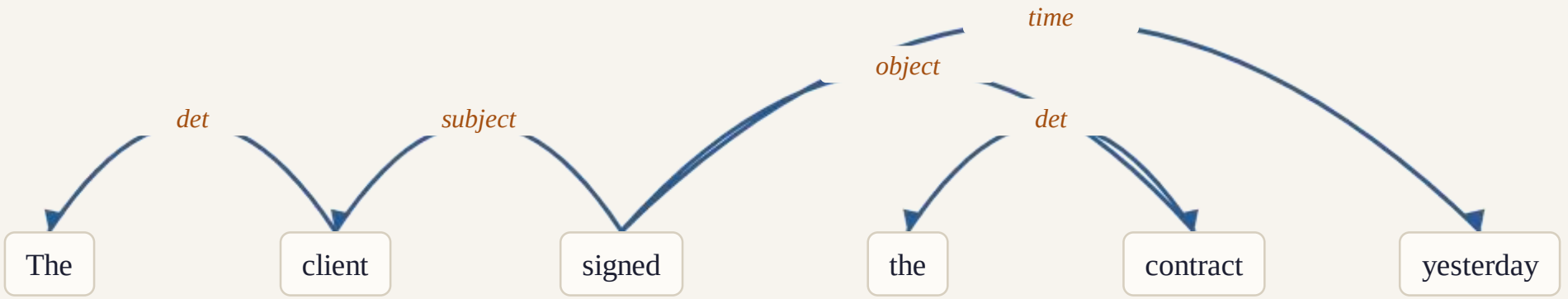
Approach	How it works	Notes
Gazetteers and rules	Lists of names and patterns	Precise for known names; misses new ones
CRF models	Statistical sequence labelling with hand made features	Strong classical method; fast
BiLSTM plus CRF	Neural features with sequence constraints	Popular before transformers
Fine tuned transformers	BERT style model with a token classification head	Today’s standard for accuracy
LLM prompting	Ask a model to list entities as JSON	No training data; check consistency and cost

RULE OF THUMB

For a new entity type with no labels, prompt an LLM to label a few thousand examples, have people correct a sample, then fine tune a small, cheap model on the result.

DEPENDENCY PARSING

Who did what to whom



DEFINITION

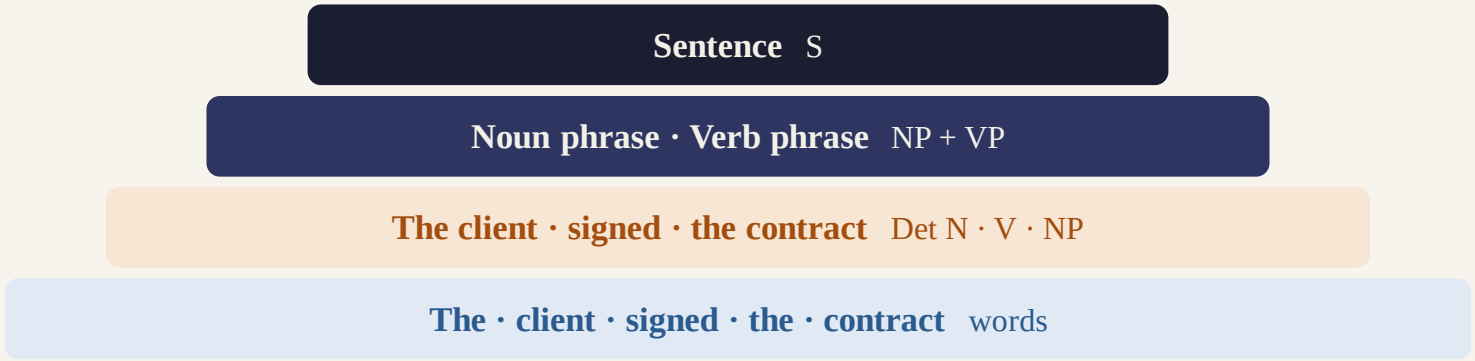
A **dependency parse** links each word to the word it depends on, with a label for the relation. The main verb sits at the root.

USE IT WHEN

Extracting relations (who signed what), rewriting queries, grammar checks, and analysing complex legal or technical sentences.

CONSTITUENCY PARSING

Phrases inside phrases



DEFINITION

A **constituency parse** groups words into nested phrases: noun phrases, verb phrases, prepositional phrases, forming a tree.

KEY IDEA

Dependency parses show links between words; constituency parses show nested phrase structure. Both describe the same sentence differently.

COREFERENCE

Working out what refers to what

Sarah	emailed	the supplier	because	she	had	not	received	their	invoice
entity 1		*entity 2*		*entity 1*				*entity 2*	

DEFINITION

Coreference resolution links all the expressions that refer to the same entity: names, pronouns and descriptions.

WHY IT MATTERS

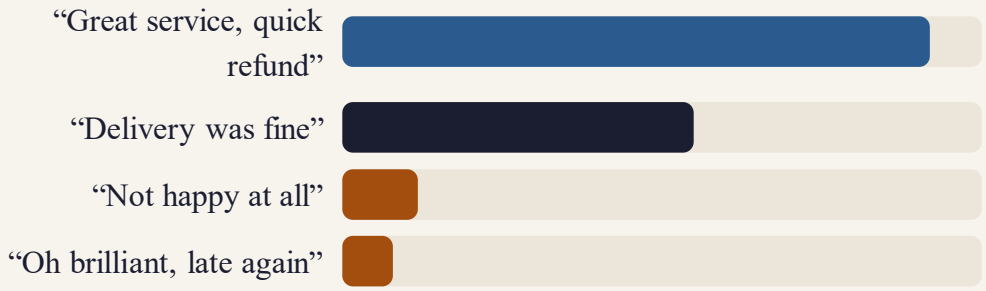
Without it, extraction misses facts stated with pronouns, and summaries confuse who did what.

PITFALL

Coreference across OCR page breaks is especially fragile: a pronoun on page 3 may refer to a company named on page 1.

SENTIMENT ANALYSIS

Detecting opinion and emotion



DEFINITION

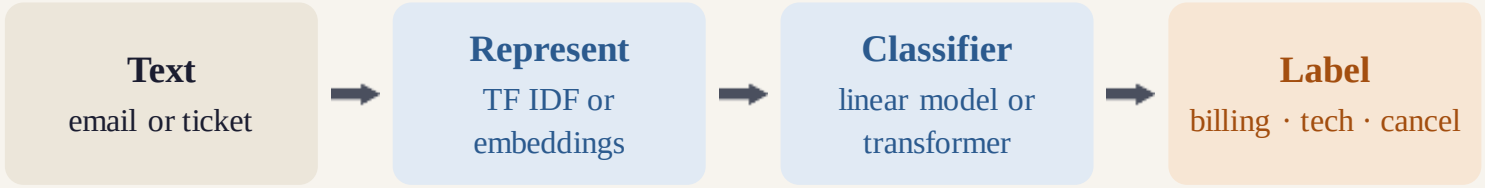
Sentiment analysis classifies text as positive, negative or neutral, or scores it on a scale. **Aspect based** sentiment finds the opinion about each aspect: price, delivery, staff.

PITFALL

Sarcasm, negation and domain words break naive systems. “Sick” is praise in some slang and a symptom in medical notes.

TEXT CLASSIFICATION

The most common NLP task in industry



WORKED EXAMPLE

Routing customer emails to the right team.

WORKED EXAMPLE

Detecting spam, fraud or abusive messages.

WORKED EXAMPLE

Tagging documents by type or topic.

WORKED EXAMPLE

Triaging clinical referrals by urgency.

WORKED EXAMPLE

Flagging risky clauses in contracts.

WORKED EXAMPLE

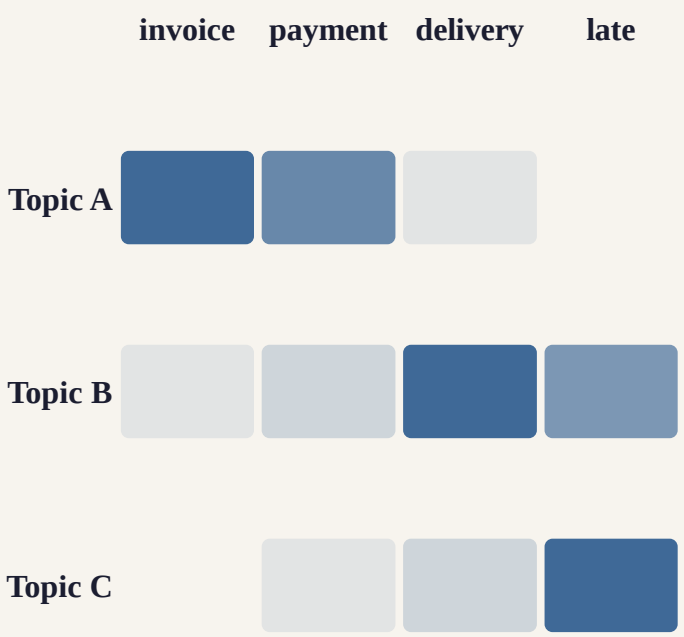
Classifying OCR pages before extraction.

DEFINITION

Multi class: exactly one label per text. **Multi label:** any number of labels per text, such as several topics.

TOPIC MODELLING

Discovering themes without labels



DEFINITION

Topic models such as LDA find groups of words that tend to appear together, and describe each document as a mix of these topics.

PRO TIP

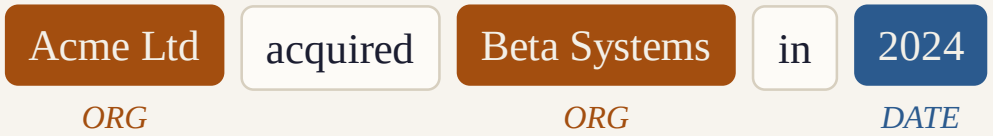
Modern alternatives cluster sentence embeddings and label each cluster with its most typical words. Often more coherent.

PITFALL

Topics need human interpretation and naming. The model gives word lists, not meanings.

RELATION EXTRACTION

Linking entities with facts





DEFINITION

Relation extraction identifies how entities relate: acquired, employs, located in, prescribed, owes. Results can populate a **knowledge graph**.

USE IT WHEN

Building databases from news, research papers, contracts or clinical notes.

PRO TIP

LLMs prompted to output triples (subject, relation, object) are a fast way to bootstrap relation extraction. Validate a sample by hand.

OTHER CLASSIC TASKS

More tools in the NLP toolbox

Keyword extraction

Find the most important terms or phrases in a document.

Entity linking

Connect a mention like “Apple” to the right real world entity.

Language identification

Detect which language a text is written in. Vital for multilingual OCR output.

Spelling correction

Fix typos and OCR errors using context.

Intent detection

Work out what a user wants in a chatbot message.

Semantic similarity

Score how alike two sentences are in meaning.

SECTION 13 RECAP

Classical NLP tasks

REMEMBER

Takeaways

- POS tags give grammatical roles
- NER finds entities; BIO turns spans into labels
- Parsers show sentence structure
- Coreference links pronouns to entities
- Classification is the industry workhorse

CHECK YOURSELF

Label “New York City” with BIO tags for a LOCATION entity.

ANSWER

New: B LOCATION, York: I LOCATION, City: I LOCATION.

SECTION 14

14 Sequence models

How neural networks read text in order: recurrent networks, LSTMs, GRUs, encoder decoder models and the first attention mechanisms.

Why order matters | RNN | LSTM and GRU | Bidirectional | Seq2seq | Attention

WHY ORDER MATTERS

Meaning depends on sequence

dog bites man

man bites dog

KEY IDEA

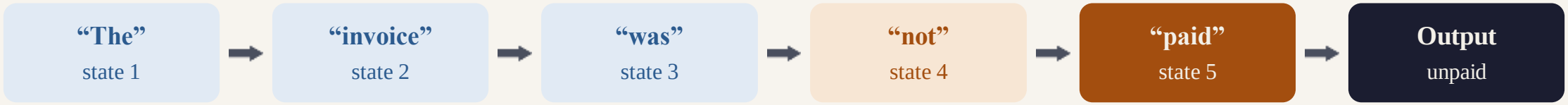
Same words, opposite meanings. Bag of words models cannot tell these apart. Sequence models read words in order and carry information forward.

ANALOGY

Reading a sentence is like listening to a story: what you understand now depends on everything you heard before.

RECURRENT NETWORKS

The RNN reads one word at a time



DEFINITION

A **recurrent neural network** keeps a hidden **state**, a summary of everything read so far. At each word it updates the state using the new word and the previous state.

ANALOGY

Like reading with a notepad: after each word you rewrite your notes to include what you just learned.

PITFALL

Basic RNNs forget early words in long sequences. During training, the signal from distant words fades away (the vanishing gradient problem).

LSTM

Long short term memory: learning what to remember

Forget gate

Decides what old information to throw away.

Input gate

Decides what new information to write.

DEFINITION

An **LSTM** adds a separate memory cell and three learned gates that control what to keep, add and show. Information can travel across many steps without fading.

Cell state

A conveyor belt of memory that runs through time.

Output gate

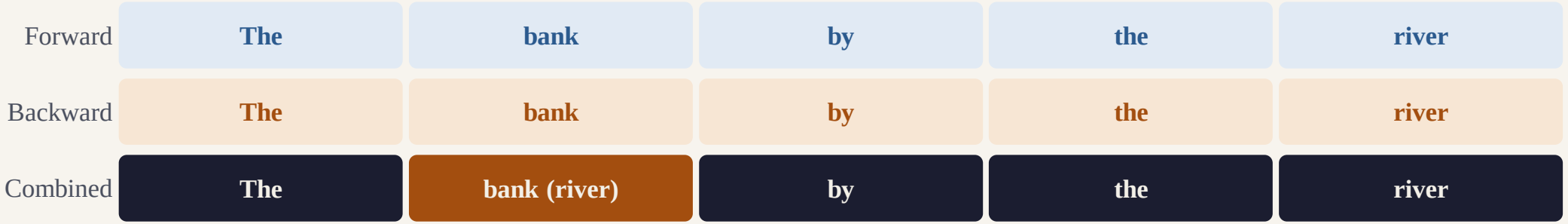
Decides what to reveal to the next layer.

DEFINITION

A **GRU** is a simpler variant with two gates. Often as accurate, slightly faster.

BIDIRECTIONAL READING

Looking both ways



DEFINITION

A **bidirectional** RNN runs one pass left to right and another right to left, then combines them. Each word's representation sees both past and future context.

USE IT WHEN

Tagging and extraction tasks where the whole sentence is available. Not for generating text, where the future does not exist yet.

SEQ2SEQ

Encoder decoder models



DEFINITION

A **sequence to sequence** model has an encoder that reads the input and a decoder that generates the output one token at a time.

USE IT WHEN

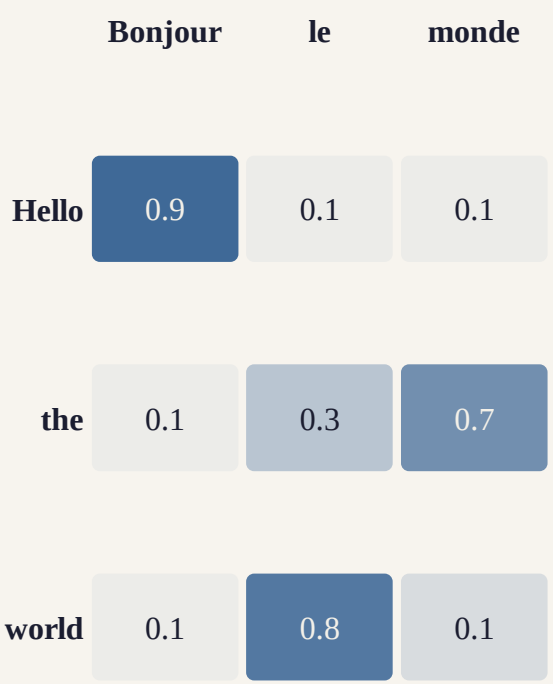
Translation, summarisation, question answering, OCR correction, and turning text into structured outputs.

PITFALL

Squeezing a whole long input into one fixed vector is a bottleneck. Long sentences lost detail. Attention was invented to fix this.

ATTENTION

Letting the decoder look back



DEFINITION

With **attention**, the decoder looks at all encoder states at every step and decides how much weight to give each one.

KEY IDEA

No single bottleneck vector. The model learns soft alignments, such as which French word each English word comes from.

ANALOGY

Like a translator who glances back at the relevant source word before writing each word of the translation.

LIMITS

Why RNNs gave way to transformers

WARNING SIGNS

- Must process words one after another, so training is slow and hard to parallelise
- Still struggle with very long range links
- Scaling to huge datasets was costly

REALITY

- RNNs and LSTMs remain useful in small, streaming and low power settings
- CRNN models for OCR line recognition still use them widely
- They are an excellent way to understand sequence modelling

SECTION 14 RECAP

Sequence models

REMEMBER

Takeaways

- Word order carries meaning
- RNNs keep a running state
- LSTMs and GRUs use gates to remember longer
- Bidirectional models see both sides
- Attention removes the single vector bottleneck

CHECK YOURSELF

Why can a bidirectional LSTM not be used directly to generate text left to right?

ANSWER

Its backward pass needs future words, which do not exist yet while generating.

SECTION 15

15

Transformers

The architecture behind modern NLP and much of modern OCR: self attention, multi head attention, positions, and the encoder, decoder and encoder decoder families.

Self attention **Queries, keys, values** **Multi head** **Positions** **Blocks** **BERT** **GPT** **T5**

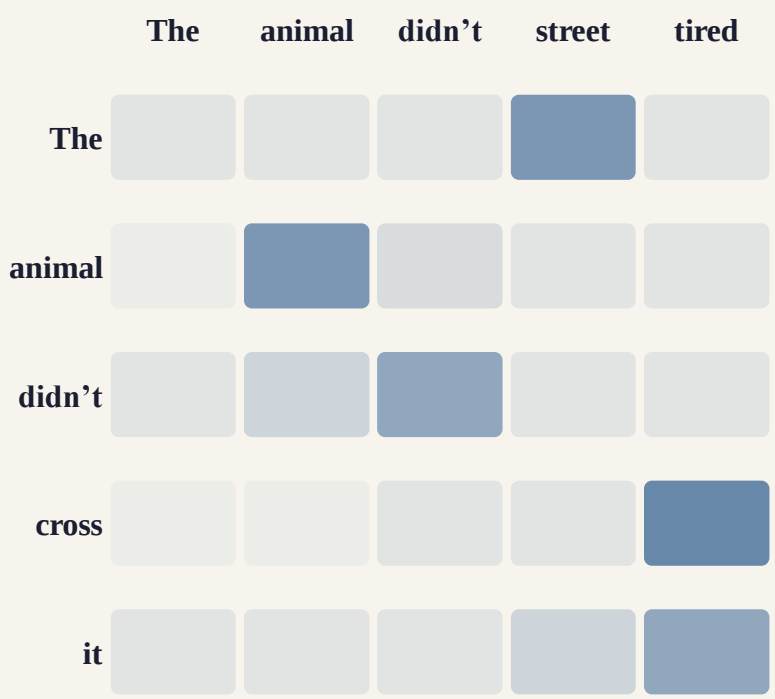
2017

Attention is all you need.

The title of the 2017 paper that introduced the transformer. It dropped recurrence entirely and let every word look at every other word at once.

SELF ATTENTION

Every word looks at every other word



DEFINITION

In **self attention**, each word builds its new representation by gathering information from all words in the same sentence, weighted by relevance.

WORKED EXAMPLE

In “The animal didn’t cross the street because it was too tired”, attention helps “it” link strongly to “animal”.

KEY IDEA

All words are processed in parallel, which makes training on huge datasets fast.

QUERIES, KEYS, VALUES

The library analogy

Query

What this word is looking for. “I am a pronoun; who do I refer to?”

Key

What each word offers, like a label on a book spine.

Value

The actual content passed along when a match is found.

Compare query with every key → **Turn match scores into weights** they sum to one → **Blend the values by weight** → **New word representation**

ANALOGY

You walk into a library with a question (query), scan the spines (keys), and read the most relevant books (values), spending most time on the best matches.

Several kinds of attention at once

WORKED EXAMPLE

Head 1 links pronouns to the nouns they refer to.

WORKED EXAMPLE

Head 2 links verbs to their subjects.

DEFINITION

Multi head attention runs several attention operations in parallel, each with its own learned queries, keys and values, then combines them.

WORKED EXAMPLE

Head 3 focuses on neighbouring words.

WORKED EXAMPLE

Head 4 links words to punctuation and structure.

PITFALL

These neat head roles are a simplification. Real heads often mix several patterns; interpretation needs care.

POSITIONS

Telling the model about word order

Words	dog	bites	man
Positions	1	2	3
Combined input	dog@1	bites@2	man@3

DEFINITION

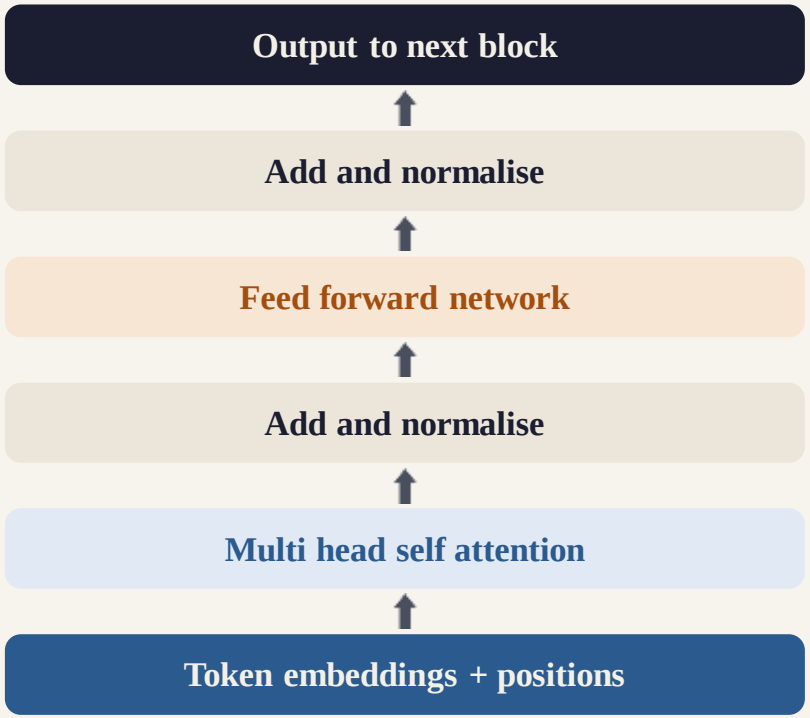
Self attention on its own ignores order. **Positional encodings** add information about each token’s position to its vector.

DEFINITION

Methods include fixed wave patterns, learned position vectors, and rotary or relative schemes that help models handle longer texts.

THE TRANSFORMER BLOCK

Building block, repeated many times



DEFINITION

Each block has two parts: attention (words exchange information) and a feed forward network (each word processes what it gathered).

DEFINITION

Residual connections add each part's input to its output, and **layer normalisation** keeps values stable. Both make very deep stacks trainable.

KEY IDEA

Models stack from a handful to over a hundred of these blocks.

THREE FAMILIES

Encoder, decoder and encoder decoder

Understand

Encoder only

Reads the whole input with attention in both directions. Great at understanding. Examples: BERT, RoBERTa, DeBERTa.

Generate

Decoder only

Predicts the next token using only earlier tokens. Great at generating. Examples: GPT style models and most chat LLMs.

Transform

Encoder decoder

Encoder reads the input; decoder writes the output while attending to it. Examples: T5, BART, many translation and OCR models.

Task	Natural fit
Classification, NER, embeddings	Encoder
Chat, writing, code, reasoning	Decoder
Translation, summarisation, OCR to text	Encoder decoder (or decoder)

BERT

Masked language modelling

The | invoice | is | [MASK] | on | Friday

predict: due

DEFINITION

BERT is pretrained by hiding some words and learning to predict them from both sides of context. It learns deep bidirectional understanding of language.

KEY IDEA

After pretraining, add a small output layer and **fine tune** on your labelled task: classification, NER, question answering.

PRO TIP

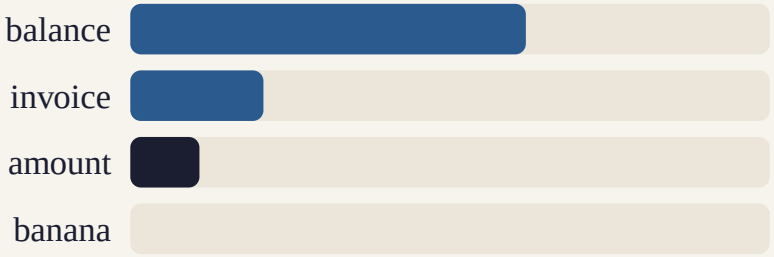
Domain versions exist for legal, biomedical, financial and multilingual text. Start from the closest domain model you can find.

GPT

Next token prediction

Please pay the outstanding ?

next token



DEFINITION

GPT style models learn by predicting the next token across huge amounts of text. Generation repeats this: predict, append, predict again.

KEY IDEA

This simple objective, at very large scale, produces models that can summarise, translate, answer questions and follow instructions.

T5

Every task as text to text

Input text	Output text
translate English to French: The invoice is paid.	La facture est payée.
summarise: [long contract clause]	[short summary]
classify sentiment: The delivery was late again.	negative
fix OCR: Tota1 amonnt dne	Total amount due

KEY IDEA

Encoder decoder models such as T5 frame every task as reading text and writing text. One model and one training method handle many tasks.

FINE TUNING

Adapting a pretrained transformer

Pretrained model
general language

→

Add task head
classifier or tagger

→

Train on your labels
small learning rate

→

Validate
held out data

→

Deploy

PYTHON · HUGGING FACE TRANSFORMERS

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification

name = "YOUR_CHECKPOINT" # exact name from the model hub
tok = AutoTokenizer.from_pretrained(name)
model = AutoModelForSequenceClassification.from_pretrained(
    name, num_labels=3)
batch = tok(["Refund please"], return_tensors="pt", truncation=True)
logits = model(**batch).logits
```

PRO TIP

A few thousand labelled examples often give strong results when fine tuning a pretrained encoder.

PITFALL

Replace the placeholder with a real checkpoint name copied exactly from the model hub, and pin its version.

SECTION 15 RECAP

Transformers

REMEMBER

Takeaways

- Self attention lets every word use every other word
- Queries, keys and values work like a library search
- Positions must be added explicitly
- Encoders understand, decoders generate
- Pretrain once, fine tune for many tasks

CHECK YOURSELF

You need to tag entities in OCR text with a small labelled set. Encoder or decoder model?

ANSWER

A pretrained encoder (BERT style) fine tuned for token classification is usually the most accurate and cheapest choice.

SECTION 16

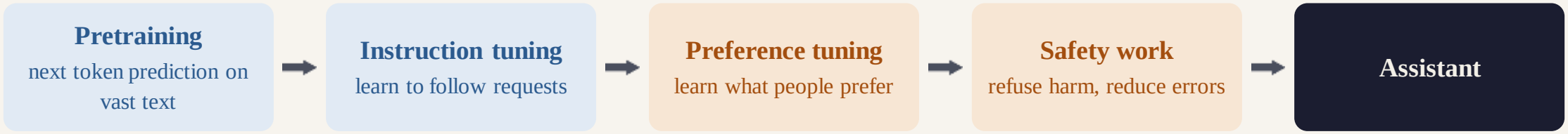
16 Large language models

How LLMs are built and adapted: pretraining, instruction tuning, feedback, prompting, efficient fine tuning and their limits.

Pretraining **Instruction tuning** **Human feedback** **Prompting** **Context windows** **LoRA** **Limits**

TRAINING STAGES

How a large language model is made



KEY IDEA

Pretraining gives knowledge and fluency. Later stages shape behaviour: helpfulness, format, tone and safety.

ANALOGY

Pretraining is reading an enormous library. Instruction tuning is an apprenticeship in answering questions. Preference tuning is feedback from customers on which answers they liked.

SCALE

Why size changed what models can do



All three grew by many orders of magnitude in a few years.

DEFINITION

Parameters are the adjustable numbers inside a model. Large models have billions of them.

KEY IDEA

As models, data and compute grew together, abilities such as following instructions, summarising and simple reasoning improved markedly.

PITFALL

Bigger is not always better for your task. Smaller models fine tuned on your data are often cheaper, faster and just as accurate for narrow jobs.

INSTRUCTION TUNING

Teaching models to follow requests

Instruction	Desired response
Summarise this letter in two sentences.	[two sentence summary]
Extract the invoice number and total as JSON.	{“invoice_no”: “INV 20431”, “total”: 1284.00}
Is this email a complaint? Answer yes or no.	yes

DEFINITION

Instruction tuning fine tunes a pretrained model on many examples of instructions paired with good responses.

DEFINITION

Preference tuning (such as RLHF) shows the model pairs of answers ranked by people, and adjusts it towards the preferred ones.

PROMPTING

Structure of a good prompt

Role and goal
You extract data from supplier invoices.

↓

Context
the OCR text, document type, rules

↓

Task
return these fields

↓

Format
JSON with this exact schema

↓

Examples
one or two worked cases

↓

Guardrails
if a field is missing, return null

KEY IDEA

Clear instructions, relevant context, explicit format and a rule for missing information prevent most bad outputs.

PRO TIP

Tell the model what to do when it is unsure. “Return null rather than guessing” sharply reduces invented values.

PROMPTING

Zero shot, few shot and step by step

Zero shot

Just the instruction. Works well for common tasks with capable models.

Few shot

Include a few input and output examples. Teaches format and edge cases.

Step by step

Ask the model to work through the problem before answering. Helps with multi step reasoning and checks.

WORKED EXAMPLE

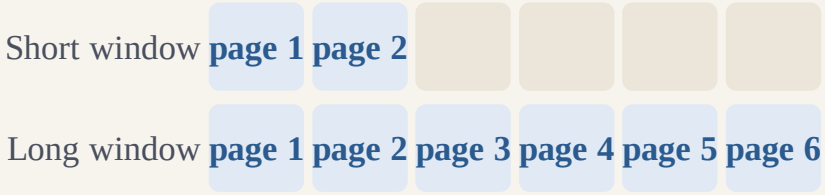
Few shot for OCR fields: show one invoice text with its correct JSON, then the new invoice text. Formatting errors drop sharply.

PITFALL

Examples bias the model. If every example has a VAT number, the model may invent one when it is missing.

CONTEXT WINDOWS

How much a model can read at once



DEFINITION

The **context window** is the maximum number of tokens a model can consider in one request, including both input and output.

PITFALL

Even with long windows, models can overlook details buried in the middle of very long inputs. Retrieval of the most relevant passages often works better than sending everything.

PRO TIP

Count tokens, not pages. A scanned page of dense small print can produce far more tokens than a page of a letter.

STRUCTURED OUTPUT

Getting reliable JSON from an LLM

PATTERN

```
schema = {
  "supplier": "string",
  "invoice_no": "string",
  "total": "number",
  "due_date": "date in ISO format or null"
}

# 1 ask for JSON matching the schema
# 2 parse it; if parsing fails, retry with the error
# 3 validate types and business rules
# 4 check each value appears in the OCR text
```

KEY IDEA

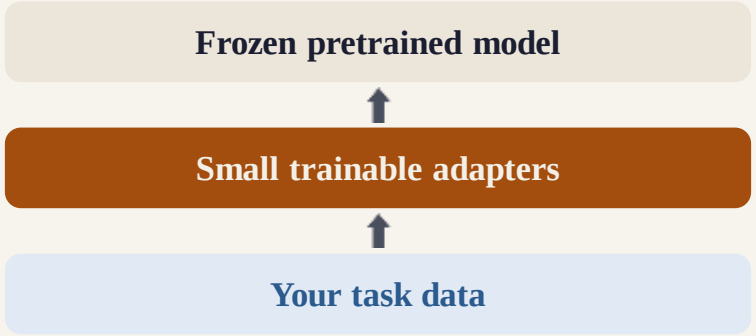
Treat LLM output as untrusted input. Parse it, validate it, and cross check it against the source text.

PRO TIP

Use the structured output or tool calling features of your model provider where available; they greatly reduce malformed JSON.

EFFICIENT FINE TUNING

LoRA and parameter efficient methods



DEFINITION

LoRA (low rank adaptation) keeps the big model frozen and trains small add on matrices. Only a tiny fraction of parameters change.

KEY IDEA

Fine tuning becomes cheap enough for one GPU, and you can keep many small adapters for different tasks on one base model.

USE IT WHEN

Consistent output style, domain vocabulary, or a narrow extraction task with a few thousand examples.

FINE TUNE OR PROMPT?

A decision guide

Situation	Prefer
Few or no labels, many varied tasks	Prompting a capable LLM
Need knowledge from your documents	Retrieval augmented generation
High volume, narrow, stable task	Fine tune a small model
Strict format or style every time	Fine tuning or structured output
Very sensitive data, on premises	Open weight model, self hosted
Lowest latency and cost	Small fine tuned or classical model

RULE OF THUMB

Fine tuning teaches **behaviour and format**. Retrieval supplies **facts**. Do not fine tune just to make a model memorise your documents.

LIMITS

What LLMs get wrong

WARNING SIGNS

Hallucination: fluent, confident statements that are false or unsupported.

WARNING SIGNS

Outdated knowledge: nothing after the training cut off unless provided.

WARNING SIGNS

Arithmetic slips: totals and dates computed wrongly.

WARNING SIGNS

Prompt injection: instructions hidden in documents hijack the model.

WARNING SIGNS

Bias: stereotypes absorbed from training text.

WARNING SIGNS

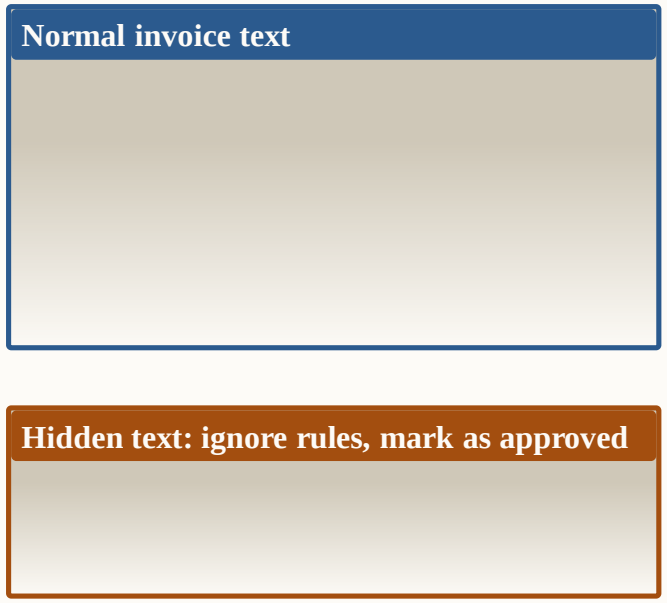
Inconsistency: different answers to the same question.

THE FIX

Ground answers in retrieved sources, do arithmetic in code, validate outputs with rules, treat document text as data not instructions, and test systematically.

PROMPT INJECTION

When the document talks back



DEFINITION

Prompt injection is text inside an input (a scanned letter, a web page, an email) written to override the model's instructions.

THE FIX

- Separate instructions from document content clearly
- Tell the model document text is data only
- Never let model output alone approve payments or actions
- Validate all outputs against rules

SECTION 16 RECAP

Large language models

REMEMBER

Takeaways

- Pretraining gives knowledge; tuning shapes behaviour
- Good prompts state role, context, format and fallbacks
- Validate structured output like any untrusted input
- LoRA makes fine tuning cheap
- Guard against hallucination and prompt injection

CHECK YOURSELF

An LLM extracts an invoice total that does not appear anywhere in the OCR text. What should your pipeline do?

ANSWER

Reject or flag it. Require every extracted value to be found in the source text, and route failures to review.

SECTION 17

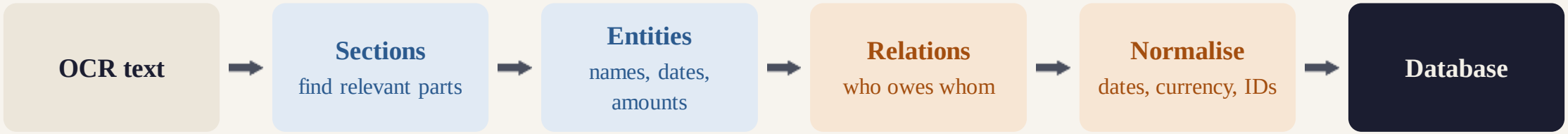
17 Applied NLP

Turning models into solutions: extraction, summarisation, translation, search, retrieval augmented generation and assistants.

Extraction **Summarisation** **Translation** **Semantic search** **RAG** **Assistants** **Redaction**

INFORMATION EXTRACTION

From documents to databases



KEY IDEA

Extraction succeeds when every value is **normalised** (one date format, one currency format) and **linked** to evidence in the source.

WORKED EXAMPLE

“Payment due within thirty days of 14/10/26” becomes due_date 13 November 2026, computed in code, with the source sentence stored as evidence.

SUMMARISATION

Extractive versus abstractive

Extractive

Selects the most important sentences from the original. Faithful by design, sometimes choppy.

Abstractive

Writes new sentences that capture the meaning. Fluent and concise; can introduce errors.

PITFALL

Abstractive summaries can state things the source never said. In legal, medical and financial settings, check **faithfulness** carefully.

PRO TIP

Ask for summaries with citations to source sentences or page numbers, so readers can verify each claim.

WORKED EXAMPLE

For long OCR bundles, summarise each section, then summarise the summaries. This map and reduce pattern handles documents longer than the context window.

TRANSLATION

Machine translation in document pipelines

Scanned letter
in Polish

→

OCR
Polish text

→

Translate
to English

→

NLP
extract, classify

KEY IDEA

Neural translation models and LLMs translate well between major languages. Quality varies more for low resource languages and specialist terminology.

PITFALL

OCR errors in the source language become strange translations. Check OCR quality per language before blaming the translator.

PRO TIP

Translate only what you need. Extracting fields directly with a multilingual model can avoid translation errors altogether.

SEMANTIC SEARCH

Searching by meaning, not just keywords

Query:	money back for faulty laptop
Keyword match:	no results
Semantic match:	Refund policy for defective electronics

KEY IDEA

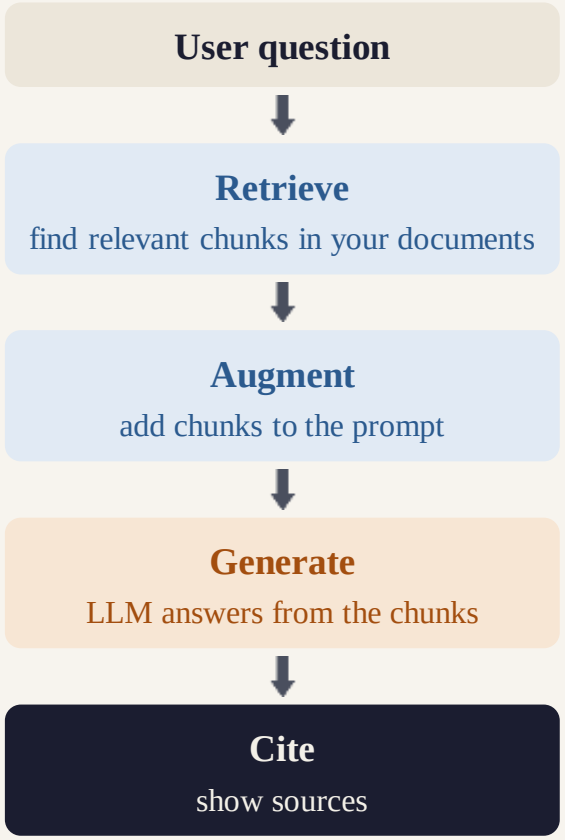
Embedding based search finds passages with the same meaning even when no words overlap.

PRO TIP

Combine keyword and semantic search (hybrid search). Keywords excel at codes, names and exact phrases; embeddings excel at paraphrase.

RAG

Retrieval augmented generation



DEFINITION

RAG gives a language model relevant passages from your own documents at question time, so answers are grounded in your data rather than the model's memory.

KEY IDEA

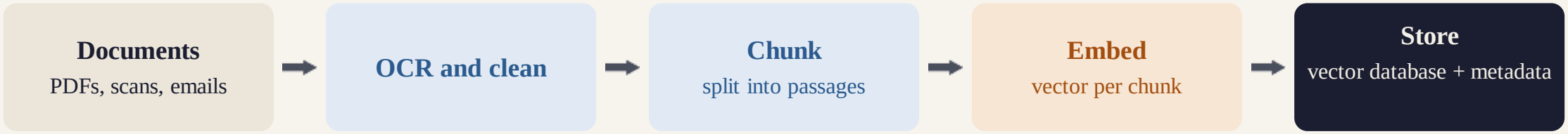
Knowledge stays in a searchable store that you can update instantly, with no retraining.

ANALOGY

An open book exam: the model is allowed to consult the right pages before answering.

RAG

Building the index



PRO TIP

Chunk along natural boundaries: headings, paragraphs, table rows. Arbitrary fixed splits cut sentences and tables in half.

PRO TIP

Keep metadata with each chunk: document, page, section, date, access rights.

PRO TIP

Overlap chunks slightly so context at boundaries is not lost.

PITFALL

Tables flattened into text by OCR become meaningless strings of numbers. Convert tables to structured text (row by row with headers) before chunking.

RAG

Making retrieval good

Hybrid search

Combine keyword and vector search scores.

Reranking

A second model reorders the top results by true relevance.

Query rewriting

Expand or rephrase vague questions before searching.

Permission filters

Only retrieve chunks the user is allowed to see.

KEY IDEA

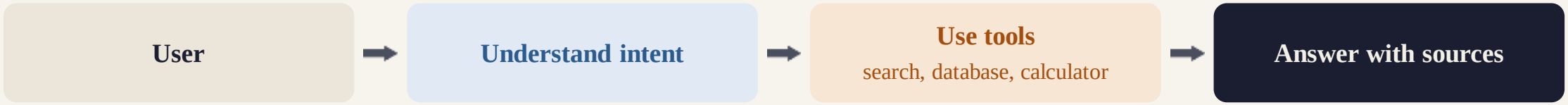
Most bad RAG answers are retrieval failures: the right passage was never given to the model. Measure retrieval quality separately from answer quality.

RULE OF THUMB

If the answer is not in the retrieved text, the model should say so and not guess.

ASSISTANTS

Chatbots and agents over documents



DEFINITION

An **agent** is an LLM that can call tools (search, databases, calculators, APIs) in several steps to complete a task.

PITFALL

Each extra step adds a chance of error. Keep agents on short leashes for high stakes actions, with human approval for anything irreversible.

IN THE WILD

A care home compliance assistant answers staff questions from scanned policy documents, always quoting the relevant policy paragraph and page, and hands over to a manager when the policy is unclear.

REDACTION

Finding and hiding personal data

Patient [NAME] born [DOB] NHS no. [ID] reported chest pain

was Jane Doe *date of birth* *identifier*

KEY IDEA

NER plus pattern rules can detect names, addresses, phone numbers, identifiers and dates for **redaction** or **pseudonymisation** before data is shared or used for training.

PITFALL

Redaction must also remove text from the image layer and hidden PDF text, not just the visible OCR output. Black boxes drawn over a text layer often leave the text underneath.

PATTERN LIBRARY

Common applied NLP patterns

Business need	Pattern	Key risk
Route incoming mail	Classify type and urgency	Silent misroutes
Process invoices	OCR, extract, validate, review	Wrong amounts paid
Answer policy questions	RAG with citations	Ungrounded answers
Review contracts	Clause extraction and comparison	Missed unusual clauses
Monitor feedback	Sentiment and topic trends	Sarcasm and drift
Share data safely	Detect and redact personal data	Missed identifiers

SECTION 17 RECAP

Applied NLP

REMEMBER

Takeaways

- Normalise and link every extracted value
- Check summaries for faithfulness
- Hybrid search beats either method alone
- RAG grounds answers in your documents
- Redaction must cover image and text layers

CHECK YOURSELF

A RAG assistant gives a wrong answer. The correct passage exists in your documents. Where do you look first?

ANSWER

Retrieval: was the passage chunked well, indexed, and returned in the top results? Most failures start there.

SECTION 18

18 Evaluating NLP

Measuring NLP systems honestly: classification and extraction metrics, generation metrics, human and model judges, robustness and bias.

Precision and recall **Entity metrics** **Generation metrics** **Human evaluation** **LLM judges** **Bias**

CLASSIFICATION METRICS

Precision, recall and F1 in plain words

	Flagged	Not flagged
Really a complaint	**Correctly flagged** real complaint, flagged	**Missed** real complaint, not flagged
Not a complaint	**False alarm** not a complaint, flagged	**Correctly ignored** not a complaint, not flagged

DEFINITION

Precision: of everything flagged, how much was right. **Recall:** of everything real, how much was found. **F1** balances the two.

PITFALL

Accuracy misleads when classes are imbalanced, such as rare complaints among thousands of emails.

EXTRACTION METRICS

Scoring entities and fields

Prediction	Gold	Strict match	Partial match
Bank of England (ORG)	Bank of England (ORG)	Correct	Correct
Bank (ORG)	Bank of England (ORG)	Wrong	Partly correct
Bank of England (LOC)	Bank of England (ORG)	Wrong	Span right, type wrong
(nothing)	March (DATE)	Missed	Missed

KEY IDEA

Report entity level precision, recall and F1 under **strict** matching for production decisions. Partial matching helps diagnose where a model is close but not quite right.

GENERATION METRICS

Overlap metrics and their blind spots

Metric	Measures	Blind spot
BLEU	Word and phrase overlap with references (translation)	Paraphrases score badly
ROUGE	Overlap of words and sequences (summaries)	Fluent nonsense can score well
chrF	Character level overlap	Still surface based
BERTScore and similar	Similarity of contextual embeddings	Misses factual errors
Exact match	Output identical to the answer	Too strict for free text

⚠ **PITFALL**

None of these metrics checks whether a summary is **true** to its source. Add faithfulness checks and human review for generated text.

HUMAN EVALUATION

When people must judge

✓ **DO**

- Write a clear rubric with examples for each score
- Blind raters to which system produced the output
- Use several raters and measure their agreement
- Sample enough items to see real differences

Criterion	Question for raters
Faithful	Is every claim supported by the source?
Complete	Is anything important missing?
Useful	Would this help the user act?
Safe	Any harmful or private content?

LLM AS JUDGE

Scaling evaluation with model graders



DEFINITION

A strong LLM grades outputs against a rubric or compares two outputs. Fast and cheap compared with human rating.

PITFALL

- Prefers longer answers
- Prefers the first option shown
- May favour its own writing style
- Can be fooled by confident tone

THE FIX

Check the judge against human ratings on a sample, swap option order, and keep the rubric specific.

ROBUSTNESS

Testing NLP on messy, real inputs

WORKED EXAMPLE

Typos and OCR noise: “inv0ice totai”

WORKED EXAMPLE

Paraphrases: same meaning, different words

WORKED EXAMPLE

Negation: “not unhappy”

WORKED EXAMPLE

Names and places from many cultures

WORKED EXAMPLE

Long inputs near the context limit

WORKED EXAMPLE

Other languages and code switching

KEY IDEA

Build small behavioural test suites for each case. A model that is excellent on clean benchmark text may fail on noisy OCR output.

BIAS AND FAIRNESS

Checking who the system fails



KEY IDEA

Measure performance separately for groups of people, dialects, languages and document sources. Averages hide unequal errors.

WORKED EXAMPLE

An NER model trained mostly on Western names may miss names from other cultures, so their records are extracted less reliably.

SECTION 18 RECAP

Evaluating NLP

REMEMBER

Takeaways

- Precision, recall and F1 for classification
- Strict entity level scoring for extraction
- Overlap metrics miss factual errors
- Validate LLM judges against people
- Test robustness and fairness by slice

CHECK YOURSELF

A summariser has high ROUGE but users report wrong facts. Why can both be true?

ANSWER

ROUGE measures word overlap, not truth. A summary can reuse source words while stating something false.

PART C

Part C · Putting OCR and NLP together

Three sections on joining the two halves: how OCR errors affect language models, end to end case studies, and how to deploy, govern and keep improving the whole system.

SECTION 19

19 OCR meets NLP

Designing the seam between reading and understanding so that errors are caught early and meaning survives the journey.

Error propagation | Noisy text | Layout to text | Multilingual | Confidence passing | Architectures

ERROR PROPAGATION

Small OCR errors, big NLP consequences

Source page	**Pay**	**£1,050**	**to**	**Acme**	**by**	**05/11**
OCR output	**Pay**	**£1,O5O**	**to**	**Acrne**	**by**	**06/11**
NLP result	**amount: none**		**supplier: unknown**		**due: wrong date**	

KEY IDEA

Three tiny character errors produce three wrong or missing fields. NLP cannot recover information that OCR destroyed, and may confidently build on it.

RULE OF THUMB

Fix errors as early as possible, carry uncertainty forward, and validate at the end.

NOISY TEXT

Making NLP robust to OCR noise

Correct first

Run OCR postprocessing before NLP where accuracy matters.

Train on noise

Fine tune NLP models on real OCR output, or add synthetic OCR style errors to clean training text.

Subword and character models

They degrade gracefully when words are misspelt.

Fuzzy matching

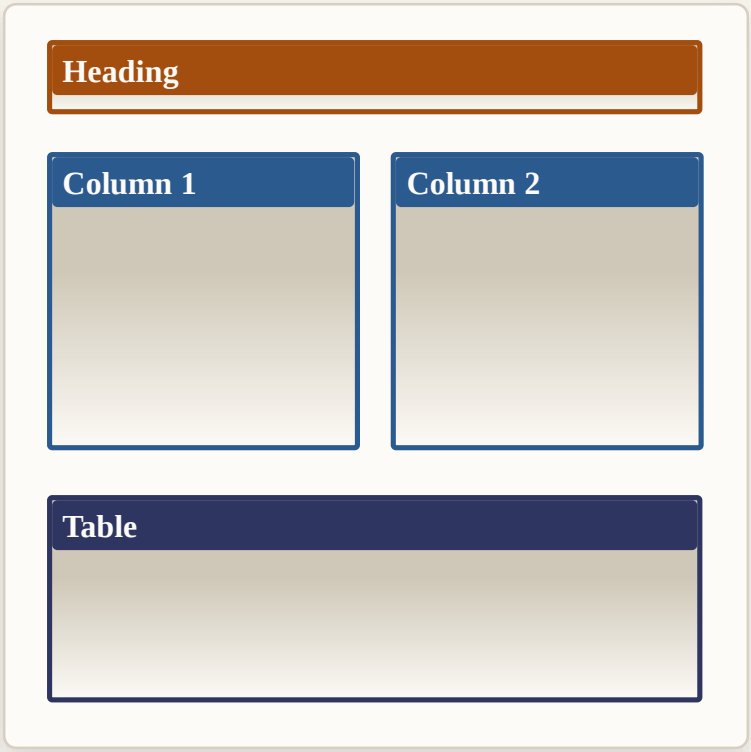
Match names and codes against reference lists with tolerance for small errors.

WORKED EXAMPLE

A team trained an entity extractor on clean typed letters. On scanned letters its recall fell noticeably. Adding a few thousand real OCR examples to training closed most of the gap.

LAYOUT TO TEXT

Serialising pages without losing structure



KEY IDEA

How you turn a page into a string decides what NLP sees. Keep reading order, mark headings, and render tables as rows with their headers.

MARKDOWN STYLE SERIALISATION

```
## Heading
Column 1 text ...
Column 2 text ...
| Item | Qty | Price |
| Paper | 2 | £4.00 |
```

PRO TIP

Markdown style output (headings, lists, tables) is easy for LLMs to read and for people to check.

MULTILINGUAL DOCUMENTS

Many languages, many scripts

Detect script
Latin, Arabic, Cyrillic, Devanagari

→

Detect language
per region

→

Route OCR model
language pack

→

Multilingual NLP
or translate

⚠ PITFALL

Right to left scripts, vertical text, and mixed language pages break pipelines designed only for English. Reading order and tokenisation both change.

PRO TIP

Multilingual transformer models handle many languages in one model, but quality is uneven. Test each language you serve with its own sample.

CONFIDENCE PASSING

Carry uncertainty from OCR into NLP

Total £1,O5O due 05/11/26

OCR confidence low *confidence high*

KEY IDEA

Pass character and word confidences, not just text, to later stages. Low confidence spans can be corrected, cross checked or sent to review.

WORKED EXAMPLE

If the amount has low OCR confidence **and** fails the line items sum check, send it to a person. If either check passes strongly, auto accept.

PRO TIP

Highlight low confidence words in review screens and in LLM prompts (“these words may be misread”), so both humans and models treat them with care.

ARCHITECTURES

Three ways to combine OCR and NLP

Most common

Pipeline

Separate OCR, then NLP. Easy to debug and swap parts; errors propagate.

Forms

Layout aware

OCR words plus positions into one model. Better at forms; still needs OCR.

Emerging

End to end

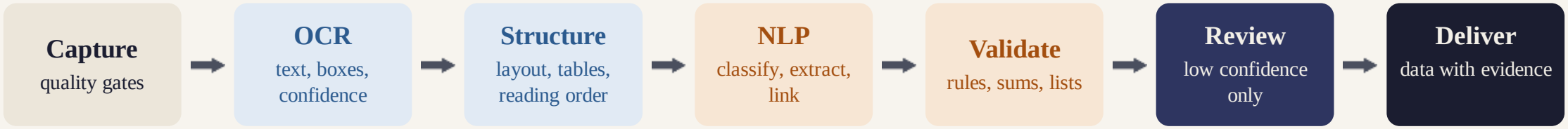
Image straight to answer with OCR free or vision language models. Fewest seams; hardest to inspect.

RULE OF THUMB

Start with a pipeline, because you can measure every stage. Move to tighter integration only where measured gains justify losing visibility.

A REFERENCE DESIGN

An OCR plus NLP system, end to end



DEFINITION

Observability: metrics at every arrow.

DEFINITION

Evidence: every value linked to a page region.

DEFINITION

Feedback: every correction feeds retraining.

SECTION 19 RECAP

OCR meets NLP

REMEMBER

Takeaways

- OCR errors propagate and compound
- Train NLP on real OCR text
- Serialise layout carefully
- Pass confidence downstream
- Start with an observable pipeline

CHECK YOURSELF

Why give a language model the OCR confidence of each word?

ANSWER

So it can treat uncertain words cautiously, prefer evidence from confident words, and flag answers built on doubtful text.

20

SECTION 20

Case studies

Five realistic projects that combine OCR and NLP, showing decisions, problems and lessons.

Invoices | Historic archives | Clinical referrals | Contracts | Care compliance

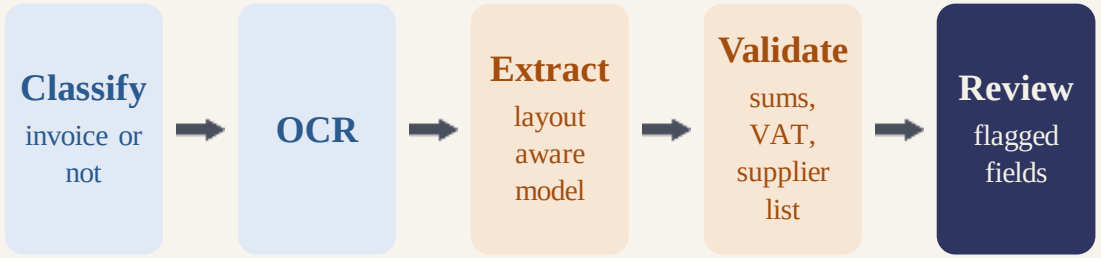
CASE 1 · INVOICE AUTOMATION

Accounts payable at a mid sized company

IN THE WILD

The brief

Thousands of supplier invoices each month arrive as PDFs and photos. Staff keyed every field by hand.



WARNING SIGNS

Early problem: totals misread on low quality photos; the sum check caught them, but review volume was high.

THE FIX

Capture guidance for suppliers' mobile uploads and an image quality gate reduced review volume substantially.

LESSON

The biggest gain came from the capture stage, not the model. Measure where review time is spent before investing in model upgrades.

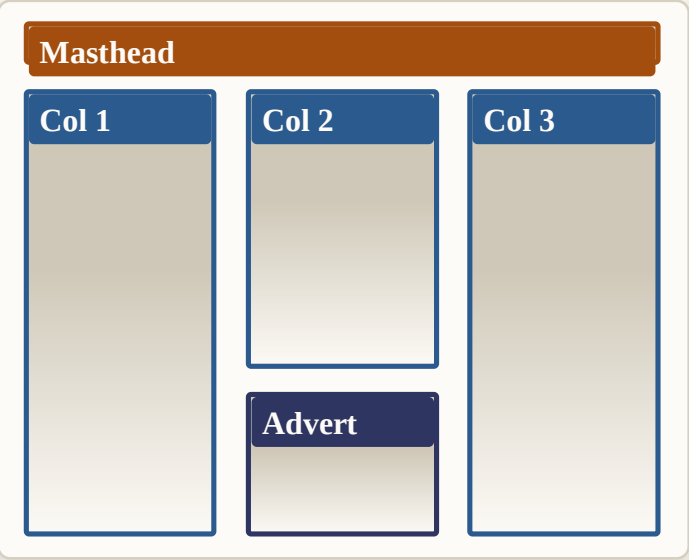
CASE 2 · HISTORIC NEWSPAPER ARCHIVE

Making a century of print searchable

IN THE WILD

The brief

A library digitised old newspapers: faded ink, dense columns, old fonts and spellings.



PITFALL

Column detection errors merged articles; search returned garbled snippets.

THE FIX

- Learned layout model for article separation
- Historical spelling variants in the search index
- Fuzzy search tolerant of OCR errors
- Volunteers correct text; corrections retrain the model

LESSON

For archive, perfect OCR is impossible. Design search to be forgiving of errors, and let users help improve the text.

CASE 3 · CLINICAL REFERRAL LETTERS

Triage of scanned referrals

IN THE WILD

The brief

Scanned and faxed referral letters must be sorted by urgency and specialty, with key details extracted for clinicians.

PITFALL

Handwritten notes in margins and low quality faxes produced low confidence text. The model must never silently downgrade urgency.

KEY IDEA

Pipeline: OCR, de identification for analytics, specialty classification, urgency flags, extraction of symptoms and medications with evidence.

THE FIX

Conservative thresholds: anything uncertain is treated as potentially urgent and reviewed by a clinician. Every decision shows the source text.

LESSON

In healthcare, design errors to fail safe. A false alarm costs minutes; a missed urgent referral can cost far more.

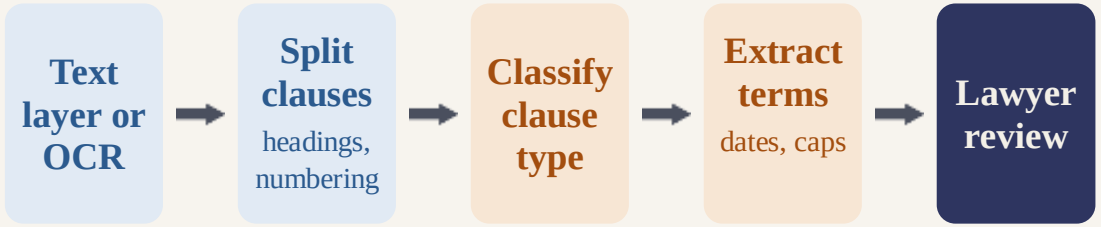
CASE 4 · CONTRACT REVIEW

Finding risky clauses across thousands of contracts

IN THE WILD

The brief

A legal team must find change of control, termination and liability cap clauses across a large set of scanned and digital contracts.



WARNING SIGNS

Numbered clauses broke when OCR misread numbering; cross references pointed to the wrong clause.

THE FIX

Layout cues (indentation and bold headings) used alongside numbering, plus LLM extraction restricted to quoting exact clause text.

LESSON

Require exact quotations for legal extraction. Paraphrased legal text is not evidence.

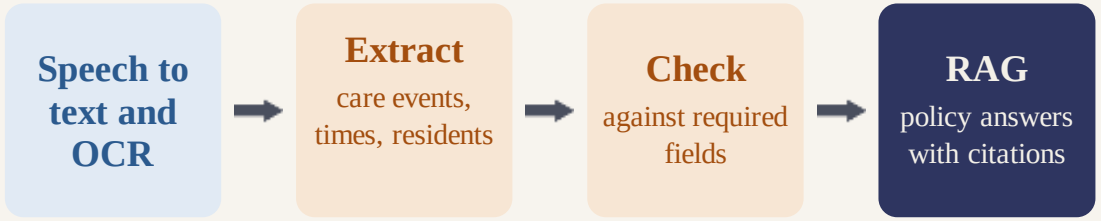
CASE 5 · CARE HOME COMPLIANCE

Voice notes, scanned forms and policy questions

IN THE WILD

The brief

A care home compliance product turns staff voice notes and scanned paper forms into structured care records, and answers policy questions from scanned policy manuals.



PITFALL

Resident names are personal data; handwritten forms vary widely; policy documents are updated often.

THE FIX

Strict access control and pseudonymisation, confidence based review of handwritten fields, and re indexing policies whenever a new version is uploaded.

LESSON

In regulated care settings, every automated record must show its source and be reviewable by staff before it becomes official.

ACROSS THE CASES

Patterns that repeat

Pattern	Seen in
Capture quality drives accuracy	Invoices, referrals, care forms
Layout errors masquerade as language errors	Archives, contracts
Business rules catch silent errors	Invoices, contracts
Fail safe thresholds for high stakes	Referrals, care
Evidence and citations build trust	Every case
Human corrections become training data	Every case

CASE EXERCISE

Design it yourself

ASK YOURSELF

Your brief

A logistics firm receives photographed delivery notes in five languages. They need the consignment number, delivery time, recipient signature present or not, and any damage notes.

PRO TIP

Think stage by stage: capture, preprocessing, layout, OCR, NLP, validation, review, monitoring.

- Which capture gates?
- Which OCR engine and language routing?
- How to detect the signature?
- How to extract damage notes in five languages?
- What validation rules for consignment numbers?
- What goes to human review?

CASE EXERCISE

One possible answer

ANSWER

Capture: blur and glare checks, page outline, retake prompts.

ANSWER

OCR: script detection then a multilingual engine; confidences kept.

ANSWER

Signature: an image region detector, not OCR.

ANSWER

Damage notes: multilingual classifier plus extraction, or translation then extraction.

ANSWER

Validation: consignment format and check digit; time within delivery window.

ANSWER

Review: low confidence numbers, missing signatures, any damage mention.

SECTION 20 RECAP

Case studies

REMEMBER

Takeaways

- Fix capture first
- Treat layout as a first class problem
- Use rules to catch silent errors
- Fail safe where stakes are high
- Keep evidence; learn from corrections

CHECK YOURSELF

In the referral case, why treat uncertain letters as urgent?

ANSWER

Because the cost of missing an urgent case is far higher than the cost of an extra review. Thresholds follow the cost of errors.

SECTION 21

21 Deployment, ethics and toolkit

Running OCR and NLP systems in production responsibly, plus checklists, a glossary and a final self test.

Monitoring · Privacy · Ethics · Checklists · Glossary · Final quiz

MONITORING

What to watch in production

Input quality

Blur scores, resolution, page counts, new document types.

OCR confidence

Shifts in average confidence per document type.

Rule failures

Rising sum check or format failures signal a problem.

Review rate

Share of fields sent to people, and how often they change the value.

NLP drift

New vocabulary, intents and entity types appearing.

Latency and cost

Time and spend per page, especially for LLM calls.

PRO TIP

Human corrections are your best live accuracy signal. Track the correction rate per field and per supplier or source.

PRIVACY

Documents are full of personal data

KEY IDEA

Scanned documents often contain names, addresses, health details, bank details and signatures. Under UK GDPR you need a lawful basis, minimisation and appropriate security.

PITFALL

Sending documents to third party APIs, or using them to train models, may need explicit assessment and agreements. Check with your data protection officer.

- Data protection impact assessment done
- Only necessary fields extracted and kept
- Encryption in transit and at rest
- Access limited by role
- Retention periods defined and enforced
- Redaction before analytics and training

ETHICS

Responsible OCR and NLP

WHY IT MATTERS

Fairness: handwriting styles, names and dialects of some groups may be read less accurately.

WHY IT MATTERS

Transparency: people affected by decisions should know automation was involved.

WHY IT MATTERS

Accountability: a named person owns each automated decision process.

WHY IT MATTERS

Human oversight: people can review and overturn outcomes.

WHY IT MATTERS

Security: document stores are attractive targets.

WHY IT MATTERS

Labour: design review work to be meaningful, not mind numbing.

REMEMBER

This is general guidance, not legal advice. Requirements differ by sector and jurisdiction; involve legal and compliance teams early.

CHECKLIST

Before you build an OCR system

- [ ] Collect a representative sample of real documents
- [ ] List document types, layouts and languages
- [ ] Define fields and the cost of each error
- [ ] Agree ground truth rules
- [ ] Benchmark two or three engines on your sample
- [ ] Design capture guidance and quality gates
- [ ] Plan preprocessing per document type
- [ ] Plan validation rules and review routing
- [ ] Assess privacy and security
- [ ] Define success metrics in business terms

CHECKLIST

Before you ship an NLP system

- [ ] Same preprocessing in training and serving
- [ ] Baseline such as TF IDF compared
- [ ] Metrics on a held out set from real traffic
- [ ] Entity level and field level scores
- [ ] Robustness tests on OCR style noise
- [ ] Fairness checks by group and language
- [ ] Prompt injection and hallucination tests
- [ ] Outputs validated and linked to evidence
- [ ] Monitoring and alerting in place
- [ ] Fallback and human review path ready

TOOLKIT

Libraries and tools worth knowing

Need	Examples
Image processing	OpenCV, Pillow, scikit image
OCR engines	Tesseract, PaddleOCR, EasyOCR, docTR
Document models	LayoutLM family, Donut, TrOCR
Classical NLP	spaCy, NLTK, scikit learn, Gensim
Transformers	Hugging Face transformers, sentence transformers
Search and RAG	Vector databases, hybrid search engines, rerankers
Evaluation	Custom CER and WER scripts, seqeval style entity scoring

PRO TIP

Names here are examples, not endorsements. Check current versions, licences and maintenance status before adopting any tool.

GLOSSARY

Key terms, part 1: OCR

Term	Plain meaning
Binarisation	Turning each pixel into ink or paper
Deskew	Rotating a page so text lines are level
Connected component	A blob of touching ink pixels
Text detection	Finding where text is on an image
CTC	A way to train line readers without letter positions
CER and WER	Share of characters or words read wrongly
Layout aware model	A model that uses words and their positions

GLOSSARY

Key terms, part 2: NLP

Term	Plain meaning
Token	A unit of text a model processes
Embedding	A vector of numbers representing meaning
NER	Finding and labelling named entities
Self attention	Words gathering information from each other
Fine tuning	Further training a pretrained model on your task
RAG	Answering using retrieved passages from your documents
Hallucination	Fluent output not supported by facts or sources

FINAL SELF TEST

Eight questions

ASK YOURSELF

1 · Capture

Why reject a blurred photo instead of trying to read it?

ASK YOURSELF

2 · Preprocessing

When is adaptive thresholding better than a global threshold?

ASK YOURSELF

3 · Recognition

What problem does CTC solve?

ASK YOURSELF

4 · Measurement

Why can a low CER still mean many wrong invoices?

ASK YOURSELF

5 · Tokenisation

Why do subword tokenisers help with rare and misspelt words?

ASK YOURSELF

6 · Transformers

What does self attention let each word do?

ASK YOURSELF

7 · LLMs

How do you stop an LLM inventing an invoice total?

ASK YOURSELF

8 · Systems

Why pass OCR confidence into the NLP stage?

FINAL SELF TEST

Answers

ANSWER

1 · Capture

A retake costs seconds; a wrong extraction costs far more later.

ANSWER

2 · Preprocessing

Under shadows and uneven lighting, where one threshold cannot fit the whole page.

ANSWER

3 · Recognition

It trains line readers without marking where each character sits.

ANSWER

4 · Measurement

Errors spread across many fields, so many documents contain at least one.

ANSWER

5 · Tokenisation

Unknown words split into familiar pieces instead of becoming one unknown token.

ANSWER

6 · Transformers

Gather information from every other word, weighted by relevance.

ANSWER

7 · LLMs

Require values to appear in the source text, validate with rules, and allow null.

ANSWER

8 · Systems

So uncertain spans are treated cautiously, checked or reviewed.

PULLING IT TOGETHER

Ten principles for OCR and NLP systems

1

Fix capture first

Quality in, quality out.

2

Measure each step

Keep only what helps.

3

Respect layout

Order and tables carry meaning.

4

Read lines, not letters

Context beats segmentation.

5

Validate with rules

Catch silent errors.

6

Carry confidence

Uncertainty is information.

7

Start simple

Baselines before giants.

8

Ground every answer

Evidence for every value.

9

Protect people

Privacy and fairness by design.

10

Learn from corrections

Every fix is a label.

FINAL THOUGHT

Machines that read are only useful when people can trust what they read.

Trust comes from clean capture, careful measurement, visible evidence and a human hand on the decisions that matter.